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Proyecto

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OCTUBRE 2022

Página

A

Diagram of a frame structure with nodes A, B, C, D, and E. The structure consists of a horizontal member BC, a vertical member BD, and a diagonal member DE. Node A is a pin support at the bottom left. Node B is a roller support at the bottom center. Node C is a roller support at the bottom right. Node D is a joint where a vertical member BD and a diagonal member DE meet. Node E is a joint where a diagonal member ED and a horizontal member EC meet. Dimensions: Horizontal distance AB = 3, BC = 4. Vertical distance BD = 4. Member ED is inclined at 45 degrees. Applied loads: A counter-clockwise moment of 1800 kg-m at node A, a vertical upward load of 600 kg at node E, and a horizontal rightward load of 2400 kg-m at node C. Internal forces are labeled: T_0 in member ED, $2T_0$ in member BD, and $3T_0$ in member BC. A handwritten note at the top right states $M_E = -3,3 \times 10^{-2}$.

1) $\sum_{D \in C} M_D = 0 \Rightarrow C_1$

2) $\Sigma H_B^{H^+} = 0 \Rightarrow H_{BC}$

3) $\Sigma M_A^+ = 0 \Rightarrow Eh$

4) $\sum_{DE} \overset{\uparrow}{M_D} = 0 \Rightarrow N_{DE}$

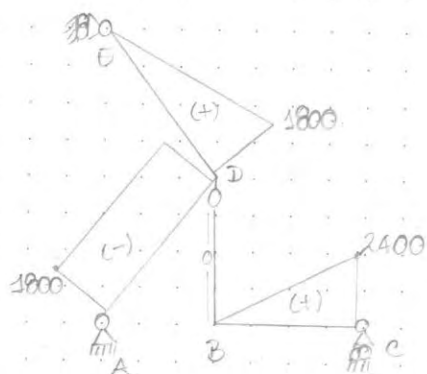
1) $\frac{J_{100}^{21}}{D_{100}} = 0 \Rightarrow 2400 + C_v \times 4 = 0 \Rightarrow C_v = -600$

$$2) \cdot \overset{+}{\underset{20}{M_B}} = 0 \Rightarrow -M_{BC} + 2400 + C_V \times 4 = 0 \Rightarrow M_{BC} = 0$$

$$3) \cdot \sum M_A^{\text{tot}} = 0 \Rightarrow 1800 + 2400 + (6 \times 7) - E_h \times 8 = 0 \Rightarrow E_h = 0$$

4) $\frac{\partial M_D}{\partial E} = 0 \Rightarrow M_{DE} - 600 \times 2 - E_H \times 4 = 0 \Rightarrow M_{DE} = 1800$

- Diagrama de Momentos:

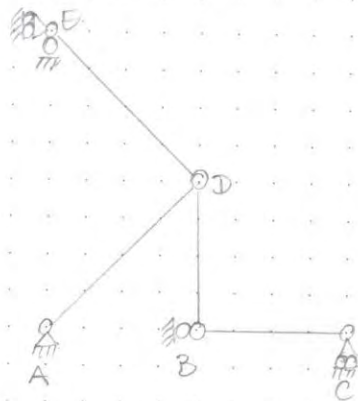


PREPARADOR: REDECOL 3 V. 2

Proyecto TEMA 2 = APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL PARA CUERPOS DEFORMABLES

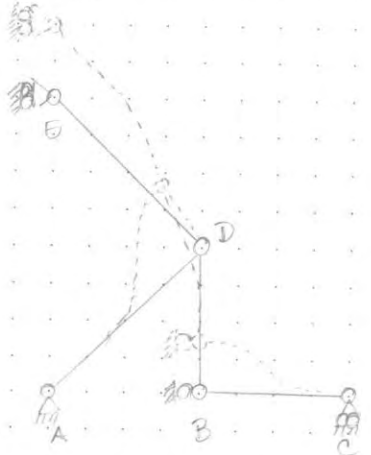
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- Imagen Cinemática

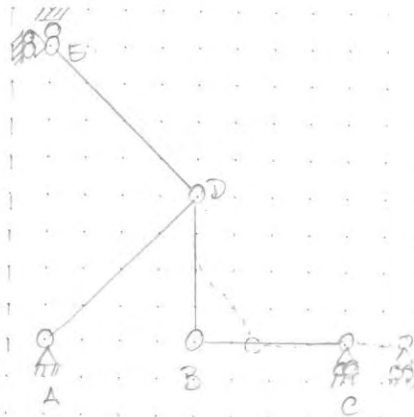


Grados de desplazabilidad = 2

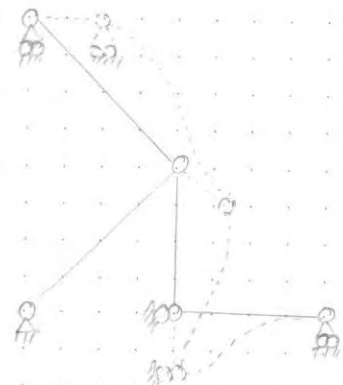
Modo I



Modo II



Modo III (U.E)



JUNTA \ MODO	A		B		C		D		E	
	u	v	u	v	u	v	u	v	u	v
I	0	0	0	1	0	0	0	1	0	1
II	0	0	0	0	0	0	0	0	0	0
III	0	0	0	1	0	0	0	1	0	0
superposición	0	0	0	1	0	0	0	1	0	1
0	0A		0B		0C		0D/0E		0E	

PREPARADOR: REYES J. VIRELLA R.

TEMA 2 = APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES

Proyecto

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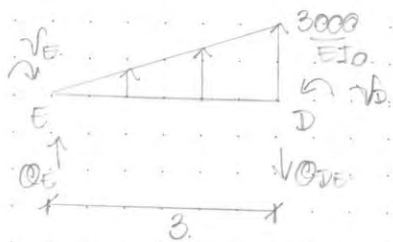
$$q_{EDE}^v = \frac{H}{EI \cos \alpha} = \frac{1800}{EI \cdot 3/5} = \frac{3000}{EI} ; q_{EDE}^h = \frac{H}{EI \cdot \sin \alpha} = \frac{1800}{EI \cdot (4/5)} = \frac{-2250}{EI}$$

$$q_{EAD}^v = \frac{H}{EI \cos \alpha} = \frac{(-1800)}{2EI \cdot 3/5} = \frac{-1500}{EI} ; q_{EAD}^h = \frac{H}{EI \cdot \sin \alpha} = \frac{(-1800)}{2EI \cdot 4/5} = \frac{-1125}{EI}$$

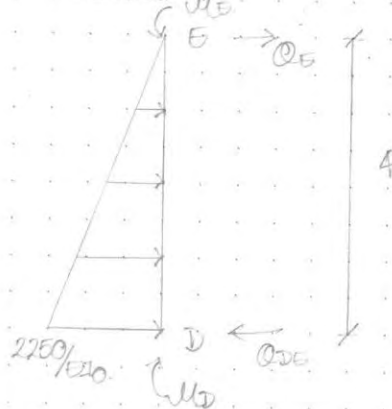
$$q_{EBC}^v = \frac{H}{3EI} = \frac{2400}{3EI} = \frac{800}{EI}$$

OVIGA DE

-Horizontal

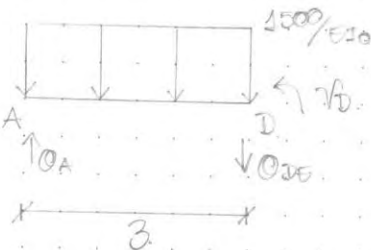


-Vertical

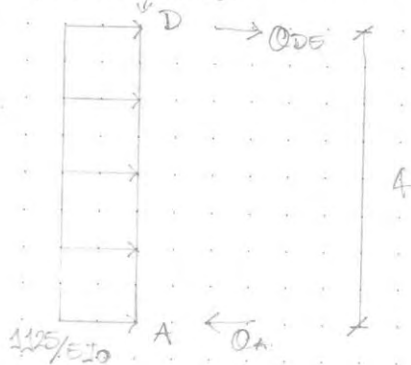


OVIGA AD

-Horizontal



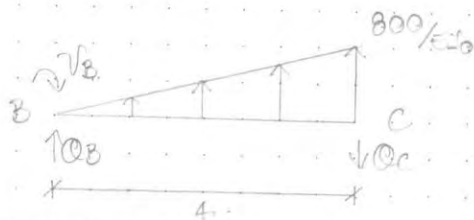
-Vertical



OVIGA BD



OVIGA BC



PREPARADOR: REYESAD I. / 18/01/12

TEMA 2 - APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUESTOS DEFORMABLES

Proyecto

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• Viga Vertical ADE

$$\sum M_E^{\text{st}} = 0 \Rightarrow M_E - Q_A \times 8 + 2250 \times \frac{4}{2} \times \frac{2}{3} \times 4 + 1125 \times 4 \times 6 = 0 \Rightarrow Q_A = \frac{9000}{E I_0}$$

• Viga Vertical AD

$$\sum F_H^{\text{st}} = 0 \Rightarrow -Q_A + Q_{DE} + \frac{1125 \times 4}{E I_0} = 0 \Rightarrow Q_{DE} = \frac{1500}{E I_0}$$

$$\sum M_D^{\text{st}} = 0 \Rightarrow M_D - Q_A \times 4 + \frac{1125 \times 4 \times 2}{E I_0} = 0 \Rightarrow M_D = \frac{27000}{E I_0}$$

• Viga Horizontal AD

$$\sum M_A^{\text{st}} = 0 \Rightarrow -Q_{DE} \times 3 - \frac{1500 \times 3 \times 3}{E I_0} + V_D = 0 \Rightarrow V_D = \frac{20250}{E I_0}$$

• Viga Horizontal DE

$$\sum M_E^{\text{st}} = 0 \Rightarrow -V_D - Q_{DE} \times 3 + V_E + \frac{3000 \times 3}{E I_0} \times \frac{2}{3} \times 3 = 0 \Rightarrow V_E = \frac{15750}{E I_0}$$

$$V_D = V_E$$

• Viga BC

$$\sum M_C^{\text{st}} = 0 \Rightarrow -Q_B \times 4 - V_E - \frac{800 \times \frac{4}{2} \times \frac{1}{3} \times 4}{E I_0} = 0 \Rightarrow Q_B = -\frac{5525,83}{E I_0}$$

$$\uparrow \sum F_V = 0 \Rightarrow Q_B - Q_C + \frac{800 \times 4}{E I_0} = 0 \Rightarrow Q_C = -\frac{3995,83}{E I_0}$$

• Viga BD

$$\sum F_H^{\text{st}} = 0 \Rightarrow Q_{DB} = Q_B = -\frac{5525,83}{E I_0}$$

$$\sum M_D^{\text{st}} = 0 \Rightarrow M_D - M_B - Q_B \times 4 = 0 \Rightarrow M_B = \frac{49383}{E I_0}$$

Presión de viento = 1400 N/m²

- Ecuaciones de la Elástica

o Viga DE

- Elástica Vertical

$$q(x) = \frac{1000}{EI_0} x$$

$$Q(x) = \frac{1}{EI_0} (500x^2)$$

$$V(x) = \frac{1}{EI_0} (500x^3 + 15750x)$$

- Elástica Horizontal

$$q(x) = \frac{1}{EI_0} (562,5x - 2250)$$

$$Q(x) = \frac{1}{EI_0} \left(\frac{562,5}{2} x^2 - 2250x + 4500 \right)$$

$$M(x) = \frac{1}{EI_0} \left(\frac{562,5}{6} x^3 - 1125x^2 + 4500x + 27000 \right)$$

o Viga AD

- Elástica Vertical

$$q(x) = -\frac{1500}{EI_0}$$

$$Q(x) = -\frac{1}{EI_0} (1500x - 9000)$$

$$V(x) = -\frac{1}{EI_0} (750x^2 - 9000x)$$

- Elástica Horizontal

$$q(x) = -\frac{1125}{EI_0}$$

$$Q(x) = \frac{1}{EI_0} (-1125x + 9000)$$

$$M(x) = \frac{1}{EI_0} \left(-\frac{1125}{2} x^2 + 9000x \right)$$

o Viga BD

$$Q(x) = -\frac{5596}{EI_0}$$

$$M(x) = \frac{1}{EI_0} (-5596x + 49383)$$

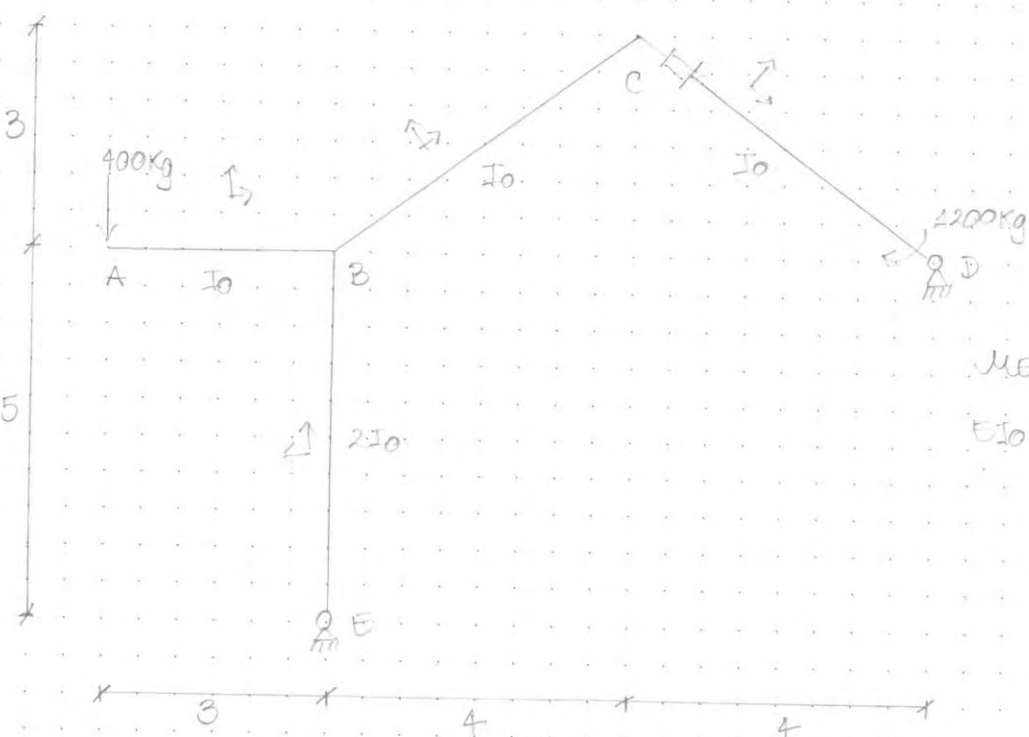
o Viga BC

$$q(x) = \frac{200}{EI_0} x$$

$$Q(x) = \frac{1}{EI_0} (100x^2 - 5596)$$

$$V(x) = \frac{1}{EI_0} \left(\frac{100}{3} x^3 - 5596x + 20250 \right)$$

2) Determine las Ecuaciones de la Elástica



$$ME = 1 \times 10^{-2} \text{ m}$$

$$EI_0 = 40^6 \text{ Kg-m}^2$$

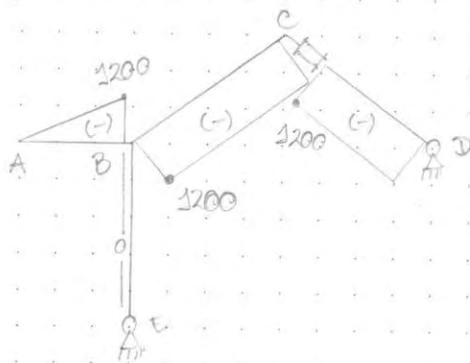
Poses para resolver la estática

$$1) \sum \overset{\curvearrowright}{M}_B = 0 \Rightarrow M_{BA} \quad 400 \times 3 + M_{BA} = 0 \Rightarrow M_{BA} = -1200 \text{ Kg}$$

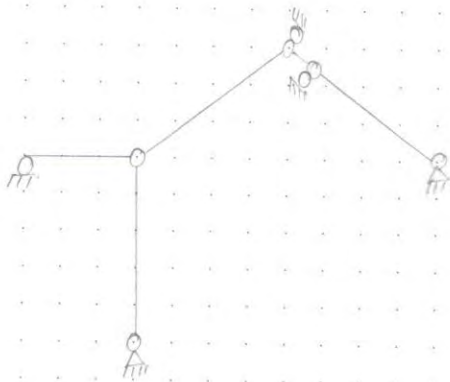
$$2) \sum \overset{\curvearrowright}{M}_D = 0 \Rightarrow E_h \quad 400 \times 3 + E_h \times 11 - 1200 = 0 \Rightarrow E_h = 0 \Rightarrow M_{BD} = 0$$

$$3) \sum \overset{\curvearrowright}{M}_E = 0 \Rightarrow M_{BE}$$

-Diagrama de Momentos:

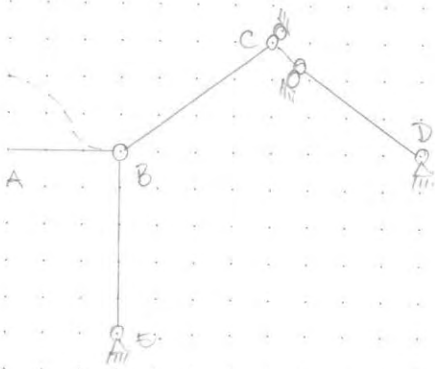


o Imagen Cinemática

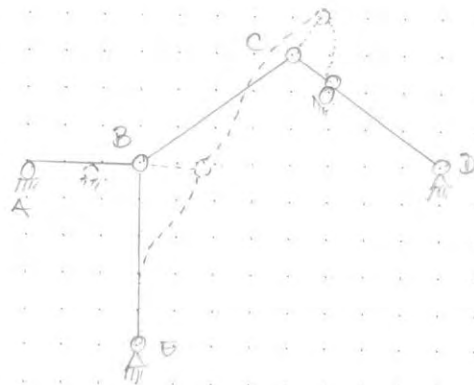


Grados de Desplazabilidad = 3

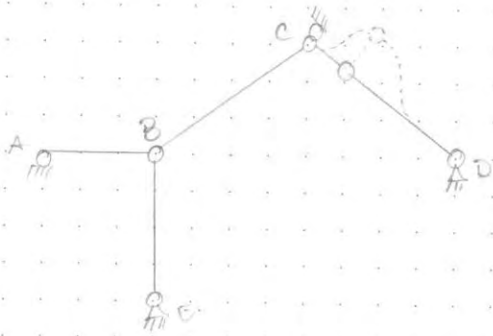
o Modo I



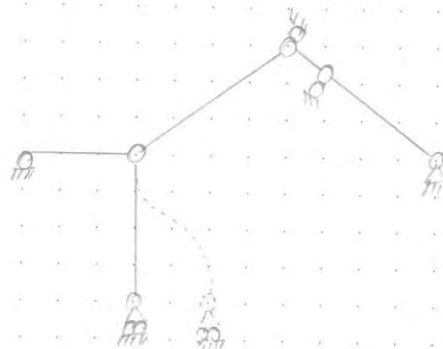
o Modo II



o Modo 3



o Modo IV. (ilic)



Proyecto TEMA 2- APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL PARA CUERPOS DEFORMABLES

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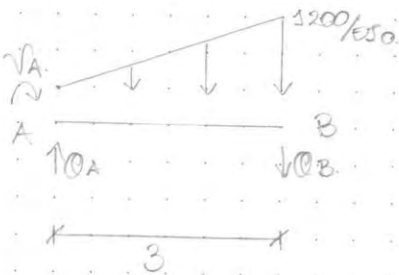
JUNTA MOV	A		B		C		D		E	
	u	v	u	v	u	v	u	v	u	v
I	0	δ_A	0	0	0	0	0	0	0	0
II	M_A	0	M_B	0	M_{CB}	δ_{CB}	0	0	0	0
III	0	0	0	0	M_{CD}	δ_{CD}	0	0	0	0
IV	0	0	0	0	0	0	0	0	M_E	0
SUPERFICIÓN	M_A	δ_A	M_B	0	M_{CB} M_{CD}	δ_{CB} δ_{CD}	0	0	M_E	0
Q	Q_A		Q_B		Q_C		Q_D		Q_E	

-Viga conjugada-

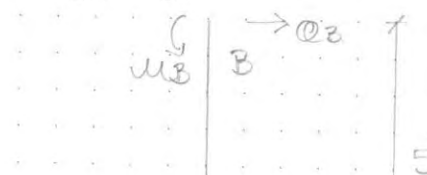
$$q_{AB}^v = -\frac{1200}{EI_0} ; q_{BC}^v = -\frac{1200}{EI_0 \cdot \frac{1}{5}} = -\frac{1500}{EI_0} ; q_{BE}^h = -\frac{1200}{EI_0 \cdot \frac{3}{5}} = -\frac{2000}{EI_0}$$

$$q_{CD}^v = -\frac{1200}{EI_0 \cdot \frac{1}{5}} = -\frac{1500}{EI_0} ; q_{CD}^h = -\frac{1200}{EI_0 \cdot (-\frac{3}{5})} = \frac{2000}{EI_0}$$

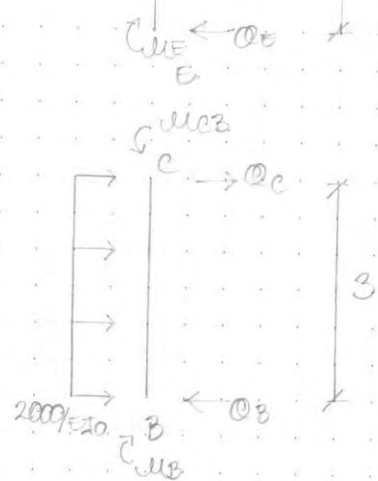
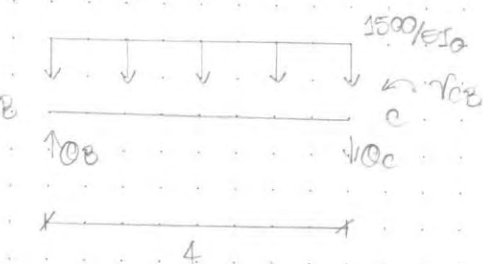
0 VIGA AB



0 VIGA BE



0 VIGA BC



PREPARADOR REVISOR J. VIGAS R

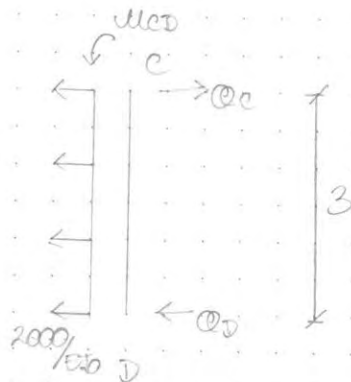
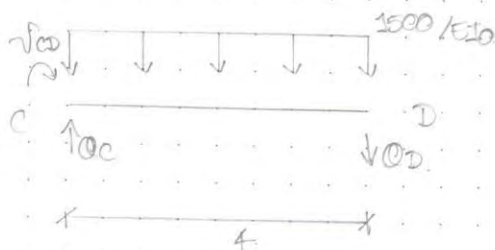
TEMA 2- APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL PARA CUERPOS DEFORMABLES.

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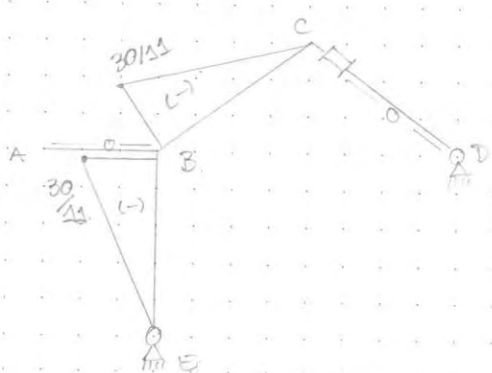
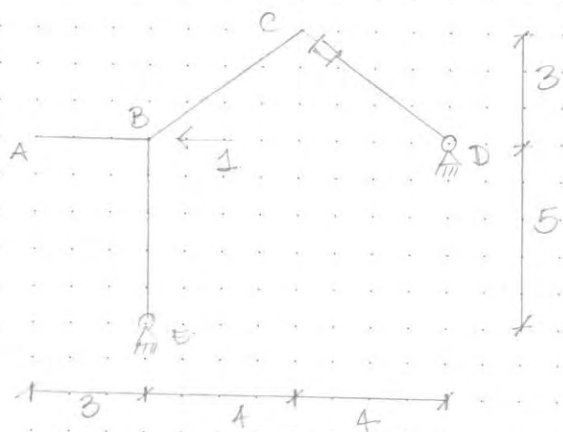
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01/16A. CD.



o Problema de Trabajo Virtual para hallar M_B
- Sistema P)



$$\sum M_P^+ = 0 \Rightarrow 1 \times 11 - 1 \times 0 = 0 \Rightarrow 11 = 6$$

$$\sum M_B^+ = 0 \Rightarrow 11 \times 5 + M_B = 0 \Rightarrow M_B = -\frac{30}{11}$$

$$M_B + \frac{0}{11} \times 1 \times 10^{-2} \times \frac{10^6}{EI_0} = -\frac{30}{11} \times \frac{(-12001 \times 5)}{2EI_0} \Rightarrow M_B = \frac{30000}{11EI_0} (\leftarrow)$$

o Viga BE

$$\sum M_E^+ = 0 \Rightarrow -M_E + M_B - Q_B \times 5 = 0 \Rightarrow Q_B = \frac{2545,45}{EI_0} = Q_E$$

o Viga AB

$$\sum F_V = 0 \Rightarrow Q_A - Q_B - \frac{1200 \times 3}{EI_0 \times 2} = 0 \Rightarrow Q_A = \frac{4345,45}{EI_0}$$

$$\sum M_A^+ = 0 \Rightarrow -V_A - Q_B \times 3 - \frac{1200 \times 3 \times \frac{3}{2} \times 3}{EI_0 \times 2} = 0 \Rightarrow V_A = -\frac{11236}{EI_0}$$

PREPARADOR: KIDDESOR J. VIGIL R.

TEMA 2 - APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES

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o Viga Horizontal BC

$$\uparrow \sum F_v = 0 \Rightarrow Q_B - Q_C - \frac{1500 \times 4}{EI_0} = 0 \Rightarrow Q_C = -\frac{3455}{EI_0}$$

$$\sum M_B = 0 \Rightarrow -Q_C \times 4 - \frac{1500 \times 4 \times 2}{EI_0} + V_{CB} = 0 \Rightarrow V_{CB} = -\frac{1820}{EI_0}$$

o Viga Vertical BC

$$\sum M_B = 0 \Rightarrow -M_B - \frac{2000 \times 3 \times \frac{3}{2}}{EI_0} - Q_C \times 3 + M_{CB} = 0 \Rightarrow M_{CB} = \frac{1362}{EI_0}$$

o Viga Horizontal CD

$$\uparrow \sum F_v = 0 \Rightarrow Q_C - Q_D - \frac{1500 \times 4}{EI_0} = 0 \Rightarrow Q_D = -\frac{2455}{EI_0}$$

$$\sum M_C = 0 \Rightarrow -V_{CD} - Q_D \times 4 - \frac{1500 \times 4 \times 2}{EI_0} = 0 \Rightarrow V_{CD} = \frac{25820}{EI_0}$$

o Viga Vertical CD

$$\sum M_D = 0 \Rightarrow -Q_C \times 3 + M_{CD} + \frac{2000 \times 3 \times \frac{3}{2}}{EI_0} = 0 \Rightarrow M_{CD} = -\frac{19365}{EI_0}$$

-Ecuaciones de la Elástica

o Viga AB

-Elástica Vertical

$$q(x) = -\frac{400}{EI_0} x$$

$$Q(x) = \frac{1}{EI_0} (-200x^2 + 4345)$$

$$V(x) = \frac{1}{EI_0} \left(-\frac{200}{3}x^3 + 4345x - 11236 \right)$$

o Viga BE

-Elástica Vertical

$$q(x) = -\frac{1500}{EI_0}$$

$$Q(x) = \frac{1}{EI_0} (-1500x + 2545)$$

$$V(x) = \frac{1}{EI_0} \left(-\frac{1500}{2}x^2 + 2545x \right)$$

o Viga BE

-Elástica Horizontal

$$Q(x) = \frac{2545}{EI_0}$$

$$M(x) = \frac{1}{EI_0} (2545x - 1000)$$

-Elástica Horizontal

$$q(x) = -\frac{2000}{EI_0}$$

$$Q(x) = \frac{1}{EI_0} (-2000x + 2545)$$

$$M(x) = \frac{1}{EI_0} \left(-1000x^2 + 2545x + \frac{30000}{2} \right)$$

Proyecto TEMA 2- APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES

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Viga CD

- Elástica Vertical

$$q(x) = -\frac{1500}{EI_0}$$

$$Q(x) = \frac{1}{EI_0} (-1500x - 3455)$$

$$V(x) = \frac{1}{EI_0} (-750x^2 - 3455x + 25820)$$

- Elástica Horizontal

$$q(x) = \frac{2000}{EI_0}$$

$$Q(x) = \frac{1}{EI_0} (2000x - 2455)$$

$$M(x) = \frac{1}{EI_0} (1000x^2 - 2455x)$$

PREPARADOR PEDRO R. J. VIREL R

TEMA 2 - APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL PARA CUERPOS DEFORMABLES

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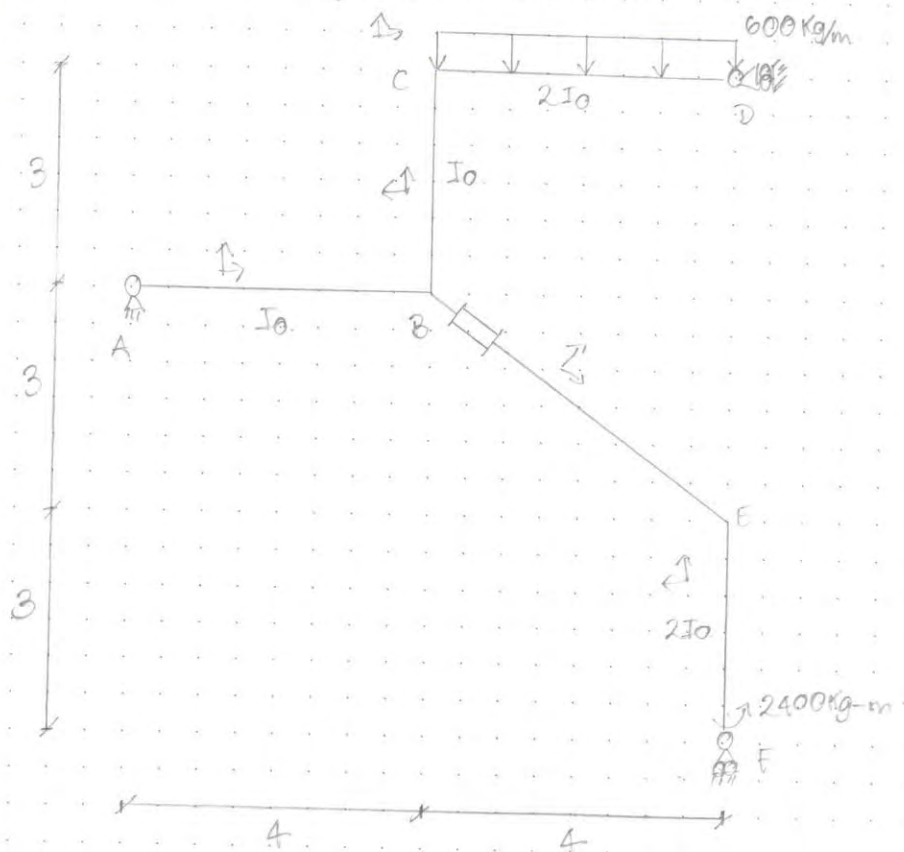
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3) Calcule el trabajo $S = 30E - 4T_D - M_F$



$$\gamma_F = -2 \times 10^{-2} \text{ m}$$

$$\Delta T_D = 15^\circ \text{C}$$

$$\alpha_t = 1 \times 10^{-5} / ^\circ \text{C}$$

$$E I_0 = 106 \text{ kg-m}^2$$

Procesos para resolver la estática:

$$1) \sum M_C = 0 \Rightarrow M_{CD}$$

$$2) \sum M_B = 0 \Rightarrow F_V$$

$$3) \sum F_V = 0 \Rightarrow A_V$$

$$4) \sum M_B = 0 \Rightarrow M_{BA}$$

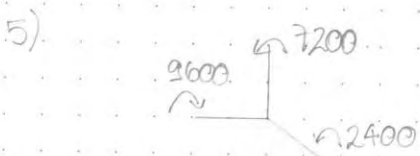
5) Equilibrio junta E.

$$1) \sum M_C = 0 \Rightarrow -M_{CD} - 600 \times 4 \times 2 = 0 \Rightarrow M_{CD} = -4800$$

$$2) \sum M_B = 0 \Rightarrow -2400 + 2400 + F_V \times 4 = 0 \Rightarrow F_V = 0$$

$$3) \sum F_V = 0 \Rightarrow F_V + A_V - 600 \times 4 = 0 \Rightarrow A_V = 2400 \text{ kg}$$

$$4) \sum M_B = 0 \Rightarrow M_{BA} - A_V \times 4 = 0 \Rightarrow M_{BA} = 9600$$



Preparador: R. RIVERA S. V. RIVERA S.

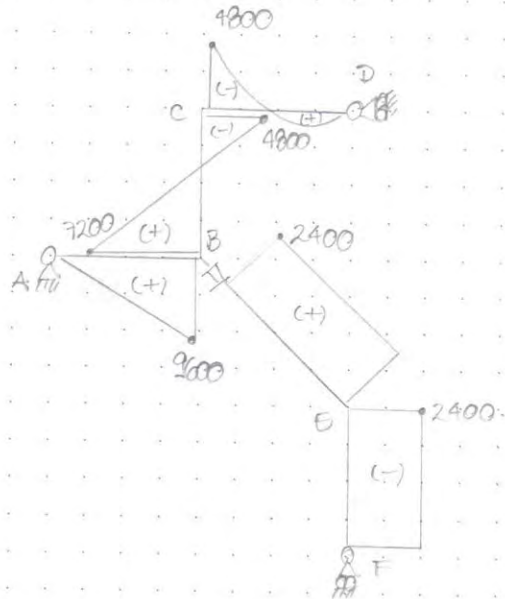
TEMA 2- APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL PARA
CUERPOS DEFORMABLES

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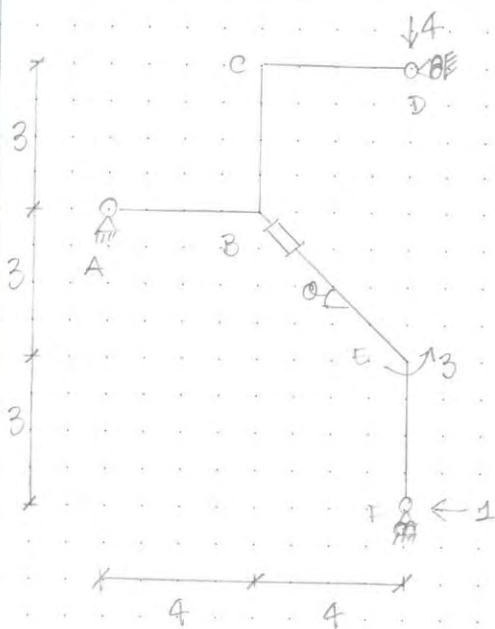
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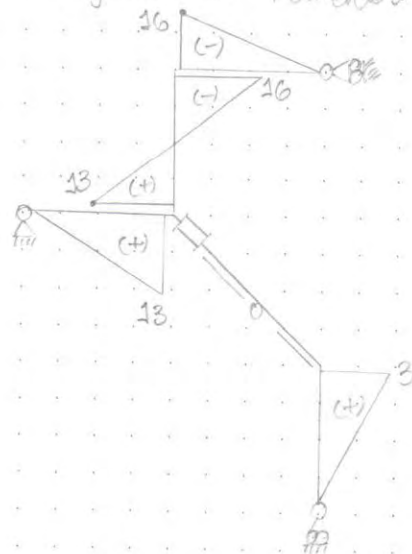
- Diagrama de Momentos sistema P.



- Sistema P'



- Diagrama de Momentos sistema P'



$$1) \sum M_C = 0 \Rightarrow -M_{CD} - 4 \times 4 = 0 \Rightarrow M_{CD} = -16$$

$$2) \sum M_E = 0 \Rightarrow 3 - 1 \times 6 + F_v \times 4 = 0 \Rightarrow F_v = 3/4$$

$$3) \sum F_v = 0 \Rightarrow A_v + F_v - 4 = 0 \Rightarrow A_v = 13/4$$

PREPARADOR REDIBUO J. VILA R.

TEMA 2 - APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES.

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$$4) \sum M_B = 0 \quad +M_{BA} - A_v \times 4 = 0 \Rightarrow M_{BA} = 13$$

$$5) \sum F_v = 0 \quad N \sin \theta + F_v = 0 \Rightarrow N = -5/4 \quad (\text{Fuerza Axial BE})$$

$$S = -2 \times 10^{-2} \times \frac{3}{4} \times \frac{10^6}{EI_0} = \frac{3 \times (-2400) \times 3}{2 \times 2EI_0} + \frac{13 \times 3600 \times 4}{3EI_0} + \frac{3}{6EI_0} \times (2 \times 13 \times 7200 + 2 \times (-16) \times (-4800)) + \dots$$

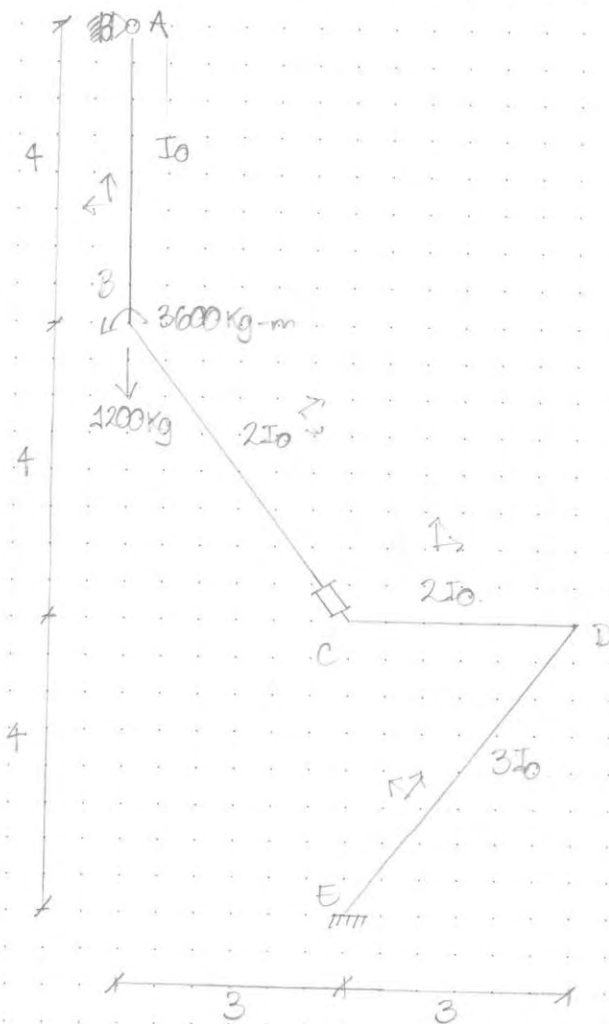
$$\dots + (-4800) \times 13 + 7200 \times (-16) + \frac{(-4800) \times (-16) \times 4}{3 \times 2EI_0} + \frac{1200 \times (-16) \times 4}{3 \times 2EI_0}$$

$$+ \frac{10^{-5} \times 10^6}{EI_0} \times 15 \times (-5/4 \times 5)$$

$$S = \frac{295062,15}{EI_0}$$

PREPARADOR: REYES J. VIGIL R.

4) Determine la Ecuación de la Elástica.



$$\theta_E = 3 \times 10^{-3} \text{ rad}$$

$$EI_0 = 10^6 \text{ kg-m}^2$$

Paseo para resolver la estática

$$1) \sum \overset{+}{M}_P = 0 \Rightarrow M_{CB}$$

$$2) \sum F_v = 0 \Rightarrow E_v$$

$$3) \sum \overset{+}{M}_P = 0 \Rightarrow M_E$$

$$4) \sum \overset{+}{M}_E = 0 \Rightarrow A_h$$

$$5) \sum \overset{+}{M}_B = 0 \Rightarrow M_{DC}$$

$$1) \sum \overset{+}{M}_P = 0 \Rightarrow 3600 - 1200 \times 3 + M_{CB} = 0 \Rightarrow M_{CB} = 0$$

$$2) \sum F_v = 0 \Rightarrow E_v - 1200 = 0 \Rightarrow E_v = 1200 \text{ kg}$$

$$3) \sum \overset{+}{M}_P = 0 \Rightarrow -E_v \times 3 - M_E = 0 \Rightarrow M_E = -3600 \text{ kg-m}$$

$$4) \sum \overset{+}{M}_E = 0 \Rightarrow -M_E + 1200 \times 3 + 3600 - A_h \times 4 = 0 \Rightarrow A_h = 900$$

$$5) \sum \overset{+}{M}_B = 0 \Rightarrow M_{DC} + 1200 \times 6 + 3600 - A_h \times 6 = 0 \Rightarrow M_{DC} = -3600$$

TEMA 2 - APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL PARA CUERPOS DEFORMABLES

Proyecto

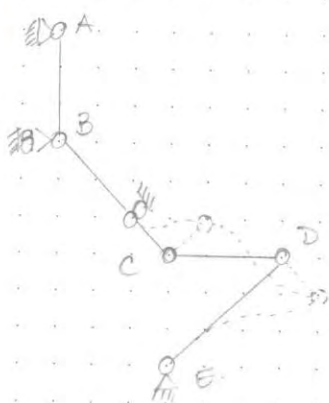
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Modo IV.



JUNTA	A		B		C		D		E	
	u	v	u	v	u	v	u	v	u	v
I	0	δ_A	0	δ_B	δ_{CB}	δ_{CD}	0	0	0	0
II	0	0	δ_B	0	δ_{CB}	δ_{CD}	0	0	0	0
III	0	0	0	0	δ_{CB}	δ_{CD}	0	0	0	0
IV	0	0	0	0	δ_{CB}	δ_{CD}	δ_D	δ_E	0	0
Desplazamiento	0	δ_A	δ_B	δ_C	δ_{CB}	δ_{CD}	δ_D	δ_E	0	0
Q	Q_A		Q_B		Q_C		Q_D		Q_E	

- Viga Conjugada

$$q_{EAB}^h = \frac{M}{EI_0} = \frac{-3600}{EI_0} ; q_{ED}^v = \frac{-3600}{2EI_0} = \frac{-1800}{EI_0}$$

$$q_{DE}^v = \frac{+/- 3600}{3EI_0 \cdot 3/5} = \frac{+/- 2000}{EI_0} ; q_{DE}^h = \frac{+/- 3600}{3EI_0 \cdot 4/5} = \frac{+/- 1500}{EI_0}$$

PREPARADOR: REDDOR J. VIERA R.

TEMA 2- APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES.

Proyecto

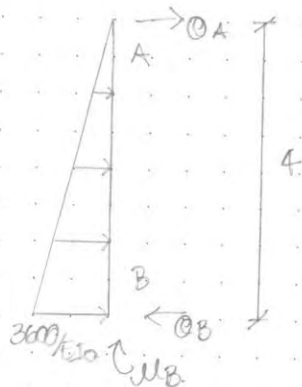
Fecha

OCTUBRE 2012

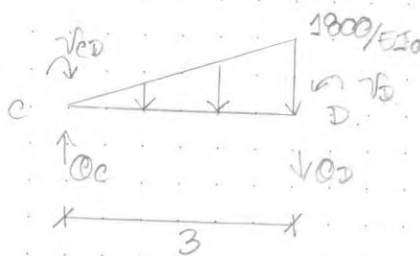
Página

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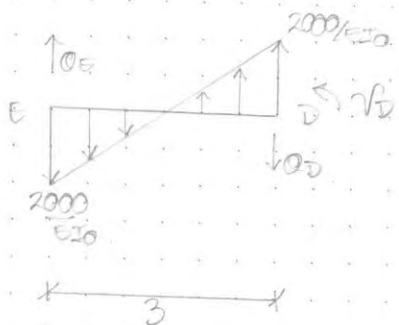
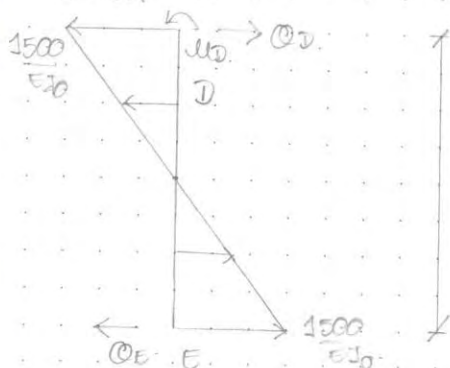
0 VIGA AB



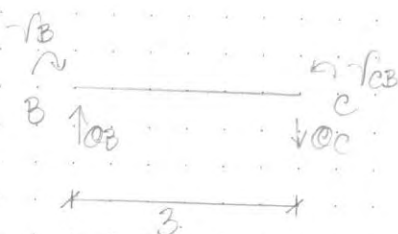
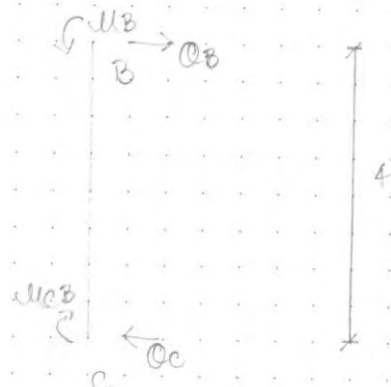
0 VIGA CD



0 VIGA DE



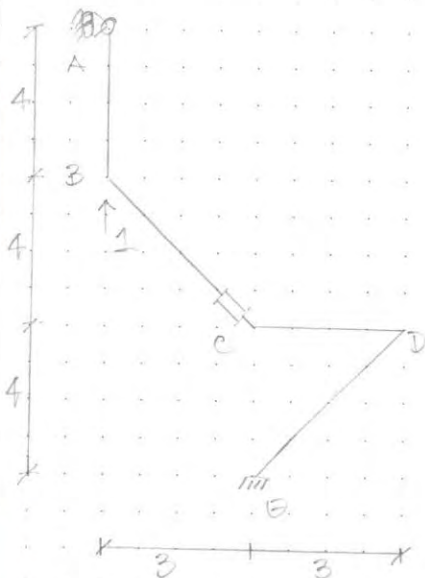
0 VIGA BC



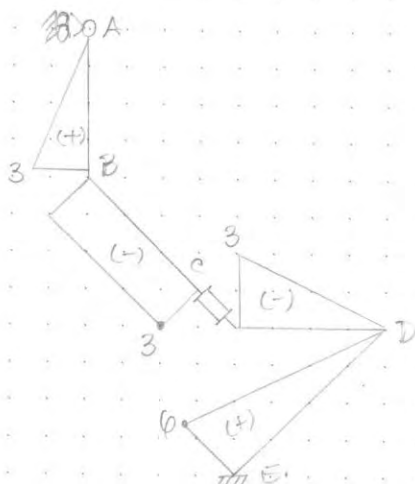
PREPARADA POR: J. VIREZ R.

- Principio de Trabajo Virtual para hallar δ_B

- Sistema P^1



- Diagrama de Momentos Sistema P^1



$$1) \sum M_P = 0 \Rightarrow M_{CB} + 1 \times 3 = 0 \Rightarrow M_{CB} = -3$$

$$2) \sum F_v = 0 \Rightarrow E_v = 1$$

$$3) \sum M_D = 0 \Rightarrow -M_E - E_v \times 3 + 3 = 0 \Rightarrow M_E = 6$$

$$4) \sum M_E = 0 \Rightarrow -M_E - 1 \times 3 - A_h \times 12 = 0 \Rightarrow A_h = -\frac{3}{4}$$

$$5) \sum M_D = 0 \Rightarrow M_{DC} - 1 \times 6 - A_h \times 8 = 0 \Rightarrow M_{DC} = 0$$

$$\delta_B = \frac{3 \times 10^{-3} \times 6 \times 10^6}{EI_0} = \frac{-3600 \times 3 \times 4}{3EI_0} + \frac{(-3600) \times (-3) \times 3}{6 \times 2EI_0} + \frac{5}{6 \times 3EI_0} \times (2 \times 6 \times (-3600)) + \frac{3600 \times 6}{6 \times 3EI_0}$$

$$\delta_B = \frac{300}{EI_0}$$

TEMA 2 - APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES

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o Viga Vertical DE

$$\sum F_H = 0 \Rightarrow -Q_E + Q_D + \frac{1500 \times 2}{EI_0} - \frac{1500 \times 2}{EI_0} = 0 \Rightarrow Q_D = \frac{3000}{EI_0}$$

$$\sum M_P = 0 \Rightarrow -Q_D \times 2 - Q_E \times 2 + 2 \times \frac{1500 \times 2}{EI_0} \times \frac{2}{3} \times 2 + M_D = 0 \Rightarrow M_D = \frac{8000}{EI_0}$$

o Viga Horizontal DE

$$\sum M_P = 0 \Rightarrow -Q_E \times \frac{3}{2} - Q_D \times 1,5 + 2 \times \frac{2000 \times 3}{EI_0} \times \frac{1}{2} \times \frac{2}{3} \times \frac{3}{2} + V_D = 0 \Rightarrow V_D = \frac{6000}{EI_0}$$

o Viga CD

$$\sum F_V = 0 \Rightarrow Q_C - Q_D - \frac{1000 \times 3}{EI_0} = 0 \Rightarrow Q_C = \frac{5700}{EI_0}$$

$$\sum M_D = 0 \Rightarrow V_D - V_{CD} - Q_C \times 3 + \frac{1000 \times 3}{EI_0} \times \frac{3}{2} = 0 \Rightarrow V_{CD} = \frac{-300}{EI_0}$$

o Viga Horizontal BC

$$\sum F_H = 0 \Rightarrow Q_B = Q_C = \frac{5700}{EI_0}$$

$$\sum M_C = 0 \Rightarrow -Q_B \times 3 - V_B + V_{CB} = 0 \Rightarrow V_{CB} = \frac{17400}{EI_0}$$

o Viga AB

$$\sum F_H = 0 \Rightarrow Q_A + \frac{3600 \times 4}{EI_0} - Q_B = 0 \Rightarrow Q_A = \frac{-1500}{EI_0}$$

$$\sum M_B = 0 \Rightarrow -M_B - Q_A \times 4 - \frac{3600 \times 4}{EI_0} \times \frac{1}{2} \times \frac{4}{3} = 0 \Rightarrow M_B = \frac{-3600}{EI_0}$$

o Viga Vertical BC

$$\sum M_B = 0 \Rightarrow -Q_C \times 4 - M_{CB} + M_B = 0 \Rightarrow M_{CB} = \frac{-26400}{EI_0}$$

-Ecuaciones de la Elástica

oViga AB

$$q(x) = \frac{300x - 3600}{EI_0}$$

$$Q(x) = \frac{1}{EI_0} (450x^2 - 3600x + 5700)$$

$$M(x) = \frac{1}{EI_0} (150x^3 - 1800x^2 + 5700x - 3600)$$

oViga DE

-Elástica horizontal

$$q(x) = \frac{1}{EI_0} (750x - 1500)$$

$$Q(x) = \frac{1}{EI_0} (375x^2 - 1500x + 3000)$$

$$M(x) = \frac{1}{EI_0} \left(\frac{250}{3} x^3 - 750x^2 + 3000x \right)$$

oViga BC:

-Elástica horizontal

$$Q(x) = \frac{5700}{EI_0}$$

$$M(x) = \frac{1}{EI_0} (5700x - 26400)$$

oViga CD

$$q(x) = \frac{-600x}{EI_0}$$

$$Q(x) = \frac{1}{EI_0} (-300x^2 + 5700)$$

$$V(x) = \frac{1}{EI_0} (-100x^3 + 5700x - 300)$$

-Elástica vertical

$$q(x) = \frac{4}{EI_0} \left(\frac{4000x}{3} - 2000 \right)$$

$$Q(x) = \frac{1}{EI_0} \left(\frac{4000}{6} x^2 - 2000x + 3000 \right)$$

$$V(x) = \frac{1}{EI_0} \left(\frac{4000}{18} x^3 - 1000x^2 + 3000x \right)$$

-Elástica vertical

$$Q(x) = \frac{5700}{EI_0}$$

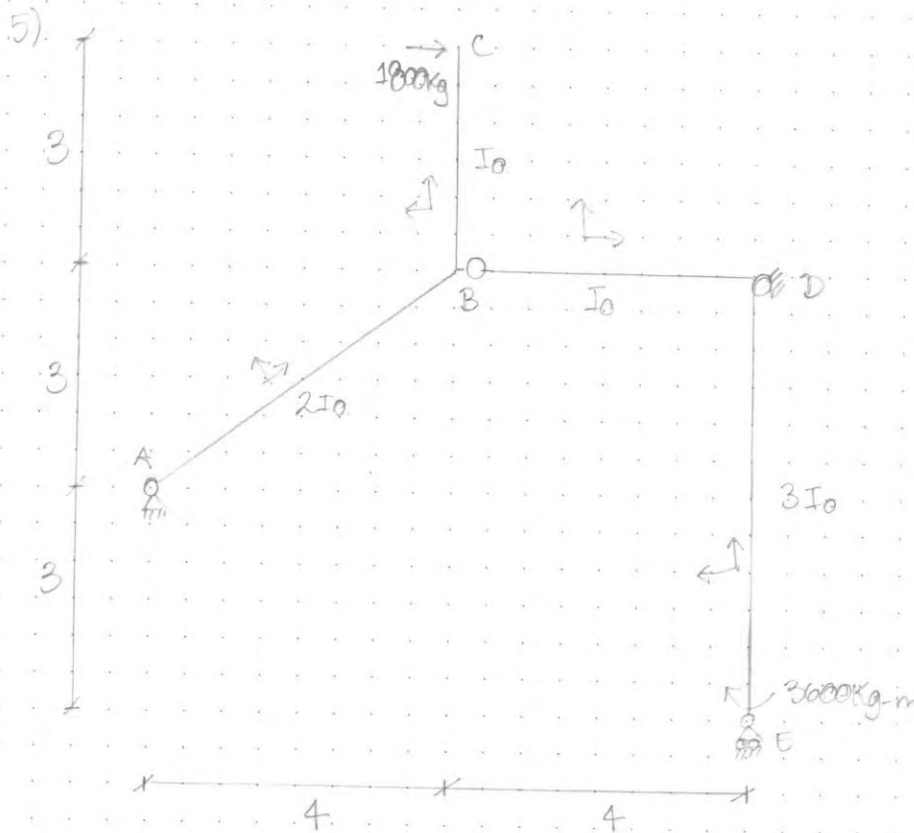
$$V(x) = \frac{1}{EI_0} (5700x + 300)$$

TEMA 2- APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES

Proyecto

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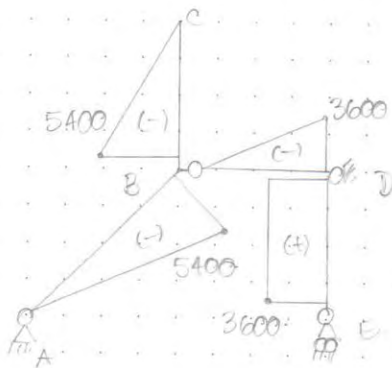


$$I_A = -4 \times 10^{-2} \text{ m}$$

$$EI_0 = 10^3 \text{ Kg} \cdot \text{m}^2$$

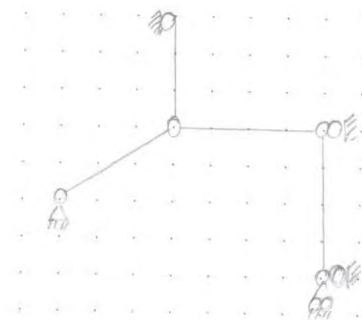
$$1) \sum \delta M_B = 0 \Rightarrow M_{BC}$$

$$1) \sum \delta M_B = 0 \Rightarrow -M_{BC} - 1800 \times 3 = 0 \Rightarrow M_{BC} = -5400$$

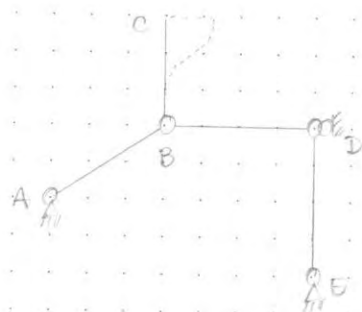


- Imagen Cinemática

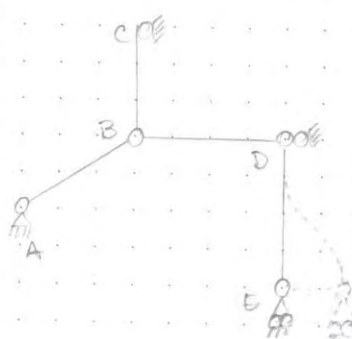
Grados de Libertad = 2



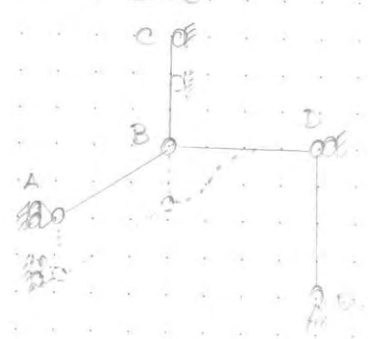
o Modo I



o Modo II



o Modo III (rA)



MODOS	A		B		C		D		E	
I	0	0	0	0	Mc	0	0	0	0	0
II	0	0	0	0	0	0	0	0	Mc	0
III (rA)	0	rA	0	rB	0	rC	0	0	0	0
SUPERPOSICIÓN	0	rA	0	rB	Mc	rC	0	0	Mc	0
Q	QA		QB		QC		QD		QE	

$$-r_A = -r_B = -r_C$$

Preparador: RODRIGUEZ S. VÍCTOR

TEMA 2- APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL PARA CUERPOS DEFORMABLES

Proyecto

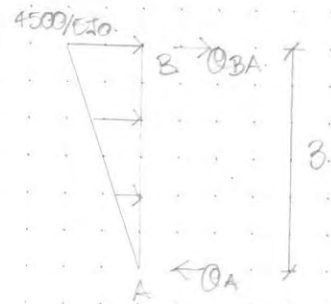
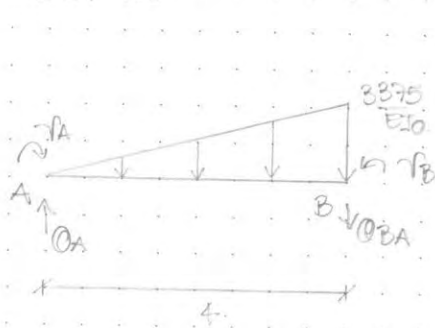
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- Viga Conjugada.

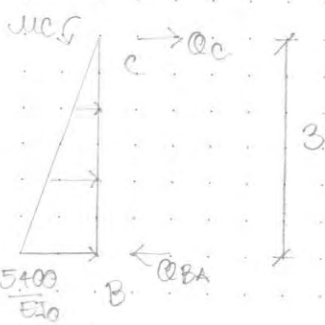
$$q_{eAB}^v = \frac{-5400}{2EI_0 \cdot \frac{4}{5}} = \frac{-3375}{EI_0} ; q_{eAB}^h = \frac{-5400}{2EI_0 \cdot \frac{3}{5}} = \frac{-4500}{EI_0} ; q_{eBC}^h = \frac{-5400}{EI_0}$$

$$q_{eBD}^v = \frac{-3600}{EI_0} ; q_{eDE}^h = \frac{3600}{3EI_0} = \frac{1200}{EI_0}$$

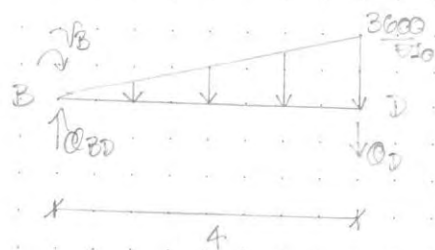
OVIGA AB



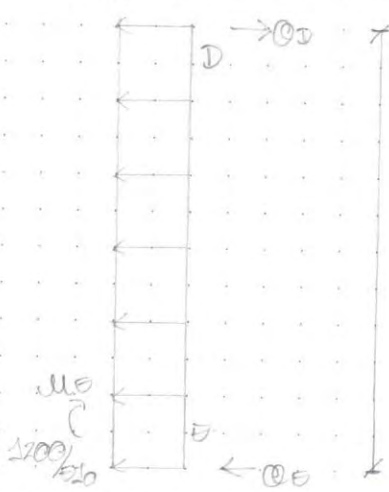
OVIGA BC



OVIGA BD



OVIGA DE



PREPARADOR REVISOR 3 VÍZEL R

TEMA 2-APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES

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o Viga Vertical AB

$$\sum \overset{\uparrow}{M}_A = 0 \Rightarrow -Q_{BA} \times 3 - \frac{4500}{E_{Ib}} \times \frac{3}{2} \times 2 = 0 \Rightarrow Q_{BA} = -\frac{4500}{E_{Ib}}$$

$$\sum \overset{+}{F}_H = 0 \Rightarrow Q_A - Q_{BA} - \frac{4500 \times 3}{E_{Ib} \cdot 2} = 0 \Rightarrow Q_A = \frac{2250}{E_{Ib}}$$

o Viga Horizontal AB

$$\sum \overset{\uparrow}{M}_A = 0 \Rightarrow -V_A - Q_{BA} \times 4 + V_B - \frac{3875}{E_{Ib}} \times \frac{4}{2} \times \frac{2}{3} \times 4 = 0 \Rightarrow V_B = -\frac{400000}{E_{Ib}} = V_A$$

o Viga BC

$$\sum \overset{+}{F}_H = 0 \Rightarrow -Q_{BA} + Q_C + \frac{5400 \times 3}{E_{Ib} \cdot 2} = 0 \Rightarrow Q_C = -\frac{12600}{E_{Ib}}$$

$$\sum \overset{\uparrow}{M}_C = 0 \Rightarrow M_C - Q_{BA} \times 3 + \frac{5400 \times 3}{E_{Ib} \cdot 2} \times 2 = 0 \Rightarrow M_C = -\frac{29700}{E_{Ib}}$$

o Viga CD

$$\sum \overset{\uparrow}{M}_B = 0 \Rightarrow -Q_D \times 4 - \frac{3600}{E_{Ib}} \times \frac{4}{2} \times \frac{2}{3} \times 4 - V_B = 0 \Rightarrow Q_D = \frac{95200}{E_{Ib}}$$

$$\sum \overset{+}{F}_V = 0 \Rightarrow Q_{CD} - Q_D - \frac{3600 \times 4}{E_{Ib} \cdot 2} = 0 \Rightarrow Q_{CD} = \frac{102400}{E_{Ib}}$$

o Viga DE

$$\sum \overset{+}{F}_H = 0 \Rightarrow Q_D - \frac{1200 \times 6}{E_{Ib}} - Q_E = 0 \Rightarrow Q_E = \frac{88000}{E_{Ib}}$$

$$\sum \overset{\uparrow}{M}_E = 0 \Rightarrow -M_E - Q_D \times 6 + \frac{1200 \times 6 \times 3}{E_{Ib}} = 0 \Rightarrow M_E = -\frac{549600}{E_{Ib}}$$

-Ecuaciones de la Elástica

o Viga AB

Elástica Vertical

$$q(x) = -\frac{84375}{E_{Ib}} \cdot x$$

$$Q(x) = \frac{1}{E_{Ib}} (-42188x^2 + 2250)$$

$$V(x) = \frac{1}{E_{Ib}} \left(-\frac{42188}{3} x^3 + 2250x - 400000 \right)$$

Elástica Horizontal

$$q(x) = -\frac{4500}{E_{Ib}} \cdot x$$

$$Q(x) = \frac{1}{E_{Ib}} (-2250x^2 + 2250)$$

$$M(x) = \frac{1}{E_{Ib}} (-2250x^3 + 2250x)$$

TEMA 2 - APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES

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o Viga BC

$$q(x) = \frac{1}{EI_0} (2000x - 5400)$$

$$Q(x) = \frac{1}{EI_0} (200x^2 - 5400x - 4500)$$

$$M(x) = \frac{1}{EI_0} (200x^3 - 2700x^2 - 4500x)$$

o Viga BD

$$q(x) = -\frac{200}{EI_0} x$$

$$Q(x) = \frac{1}{EI_0} (-450x^2 + 102400)$$

$$V(x) = \frac{1}{EI_0} (-150x^3 + 102400x - 400.000)$$

o Viga DE

$$q(x) = \frac{1200}{EI_0}$$

$$Q(x) = \frac{1}{EI_0} (1200x + 88000)$$

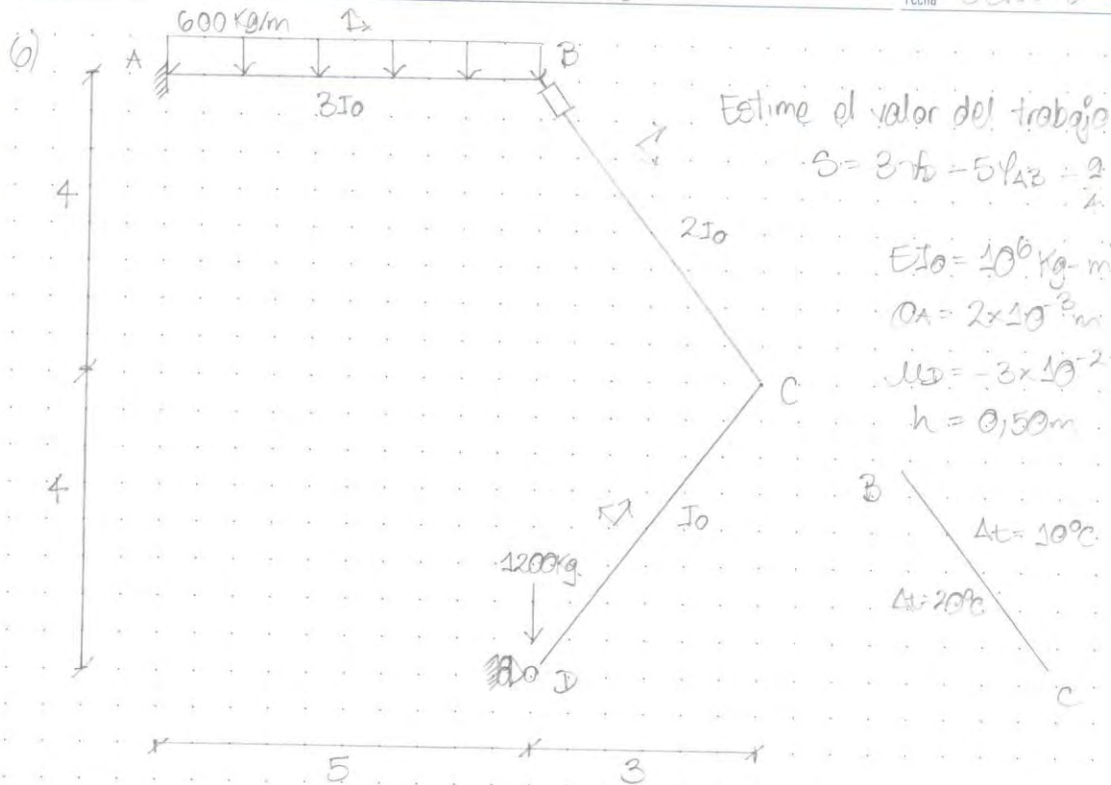
$$M(x) = \frac{1}{EI_0} (600x^2 + 88000x - 542600)$$

PREPARADOR: REDONDO J. VÍCTOR R.

TEMA 2-APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES

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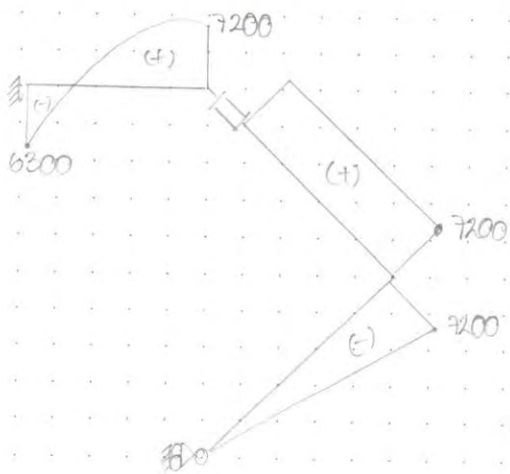


$$1) \sum M_B = 0 \Rightarrow 1200 \times 6 - M_B = 0 \Rightarrow M_B = 7200 \text{ kg} \cdot \text{m}$$

$$2) \sum M_D = 0 \Rightarrow -M_B + D_h \times 8 = 0 \Rightarrow D_h = 900 \text{ kg}$$

$$3) \sum M_A = 0 \Rightarrow -M_A - 600 \times 5 \times \frac{5}{2} - 1200 \times 5 + D_h \times 8 = 0 \Rightarrow M_A = -6300 \text{ kg} \cdot \text{m}$$

Diagrama de Momentos Sistema P:

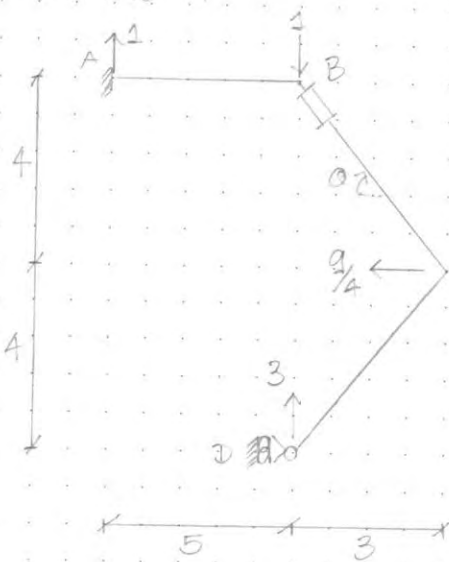


Proyecto TEMA 2-APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL PARA CUERPOS DEFORMABLES

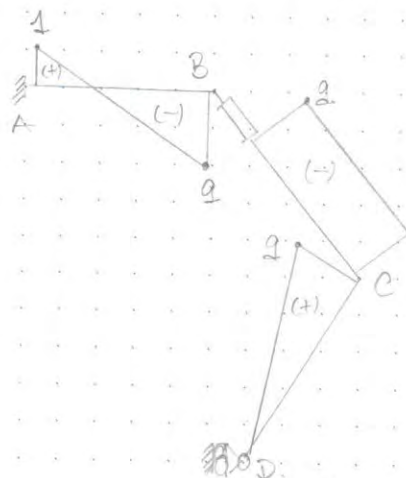
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Sistema P'



-Diagrama de Momentos Sistema P'



$$1) \sum \overset{\curvearrowright}{M}_B = 0 \Rightarrow -M_B - 3 \times 6 + 2 \times 4 = 0 \Rightarrow M_B = -2 \text{ Kg-m}$$

$$2) \sum \overset{\curvearrowright}{M}_B = 0 \Rightarrow -M_B - 2 \times 4 + D_h \times 8 = 0 \Rightarrow D_h = 0$$

$$3) \sum \overset{\curvearrowright}{M}_A = 0 \Rightarrow -M_A - 1 \times 5 - 2 \times 4 + 3 \times 5 + D_h \times 8 = 0 \Rightarrow M_A = 1 \text{ Kg-m}$$

$$4) \sum F_v = 0 \Rightarrow N_{BC} + 3 = 0 \Rightarrow N = -3.75 \text{ Kg (Fuerza axial BC)}$$

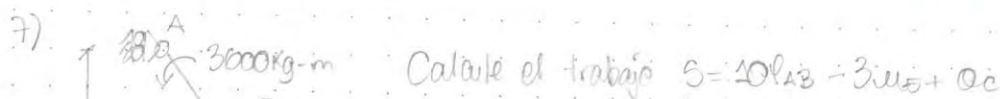
$$S = 2 \times 10^{-3} \times 1 \times \frac{10^6}{EI_0} = \frac{(-7200) \times 2 \times 5}{3EI_0} + \frac{(-7200) \times (-2) \times 5}{2EI_0} + \frac{2}{6 \times 2EI_0} \times [2 \times (1) \times (-6300) + 2 \times (17200) \times (1 \cdot 9)] +$$

$$1 \times 7200 + (-2) \times (-6300)] + \frac{1 \times 1875 \times 5}{3 \times 2EI_0} + \frac{(-2) \times 1875 \times 5}{3 \times 2EI_0} + 10^{-5} \times \frac{10^6}{EI_0} \times 15 \times [3.75 \times 5]$$

$$+ \frac{10^{-5} \times 10^6}{EI_0} \times 10 \times (-2) \times 5 \times \frac{1}{0.150}$$

$$S = 302825,03 / EI_0$$

PREPARADOR REDDOR J VIGIL R

7)  Calcule el trabajo $S = 10I_{AB} - 3I_{BC} + 0C$

$$EI_0 = 10^6 \text{ kg-m}^2$$

$$\alpha = 1 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$$

$$h_{BC} = 0,50 \text{ m}$$

$$M_D = -2 \times 10^2 \text{ m}$$

$$\frac{20^\circ}{10^\circ} \quad B \quad C$$

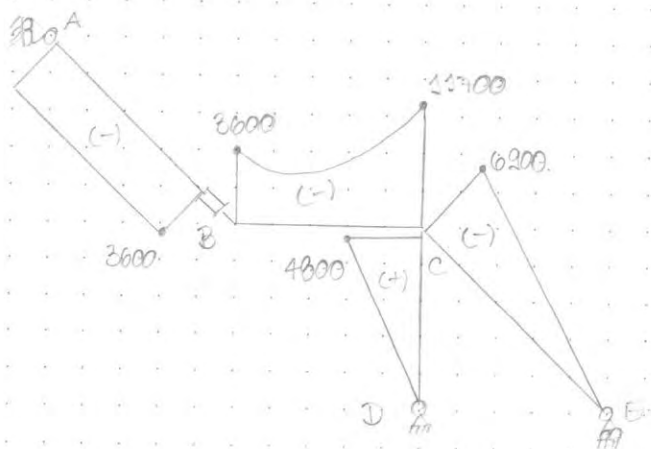
$$1) \sum M_B = 0 \Rightarrow -A_h \times 4 = 0 \Rightarrow A_h = 0$$

$$2) \sum F_H = 0 \Rightarrow A_h + D_h + 1200 = 0 \Rightarrow D_h = -1200 \text{ kg}$$

$$3) \sum M_C = 0 \Rightarrow M_{CD} + D_h \times 4 = 0 \Rightarrow M_{CD} = 4800 \text{ kg-m}$$

$$4) \sum M_A = 0 \Rightarrow M_{AB} + 600 \times 3 \times \frac{3}{2} + 3600 - A_h \times 4 + 1800 \times 2 = 0 \Rightarrow M_{AB} = -11700 \text{ kg-m}$$

Diagrama de Momentos Sistema P.



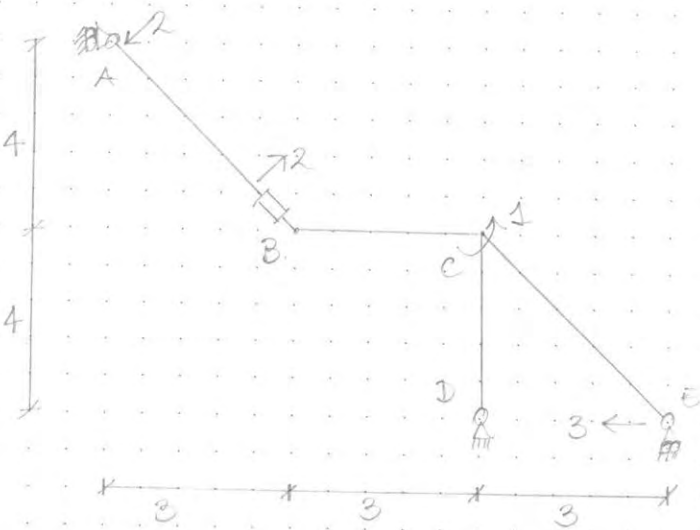
TEMA 2-APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
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- Sistema P



$$1) \sum M_B^{\pm} = 0 \Rightarrow 2 \times 5 + M_{BA} - A_h \times 4 = 0 \Rightarrow A_h = 0$$

$$2) \sum F_h^{\pm} = 0 \Rightarrow A_h + D_h - 3 = 0 \Rightarrow D_h = 3 \text{ kg}$$

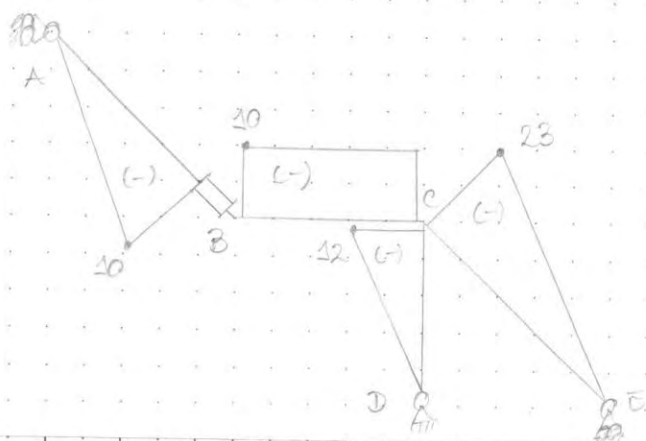
$$3) \sum M_C^{\pm} = 0 \Rightarrow M_{CD} + D_h \times 4 = 0 \Rightarrow M_{CD} = -12 \text{ kg-m}$$

$$4) \sum M_E^{\pm} = 0 \Rightarrow 1 + E_v \times 3 - 2 \times 5 + 2 \times 10 - A_h \times 8 = 0 \Rightarrow E_v = -\frac{11}{3} \text{ kg}$$

$$5) \sum M_C^{\pm} = 0 \Rightarrow E_v \times 3 - 3 \times 4 + M_{CE} = 0 \Rightarrow M_{CE} = -23 \text{ kg-m}$$

$$6) \sum F_h^{\pm} = 0 \Rightarrow A_h + M_{BC} = 0 \Rightarrow M_{BC} = 0 \text{ (Fuerza Axial BC)}$$

- Diagrama de Momentos Sistema P



Profesor: P. R. R.

Proyecto TEMA 2- APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES

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$$S = 2 \times 10^{-2} \times 3 \times \frac{10^6}{EI_0} = \frac{(-3600) \times (-10) \times 5}{2 \times 2EI_0} + \frac{4000 \times (-12) \times 4}{3EI_0} + \frac{(-6900) \times (-13) \times 5}{3EI_0}$$

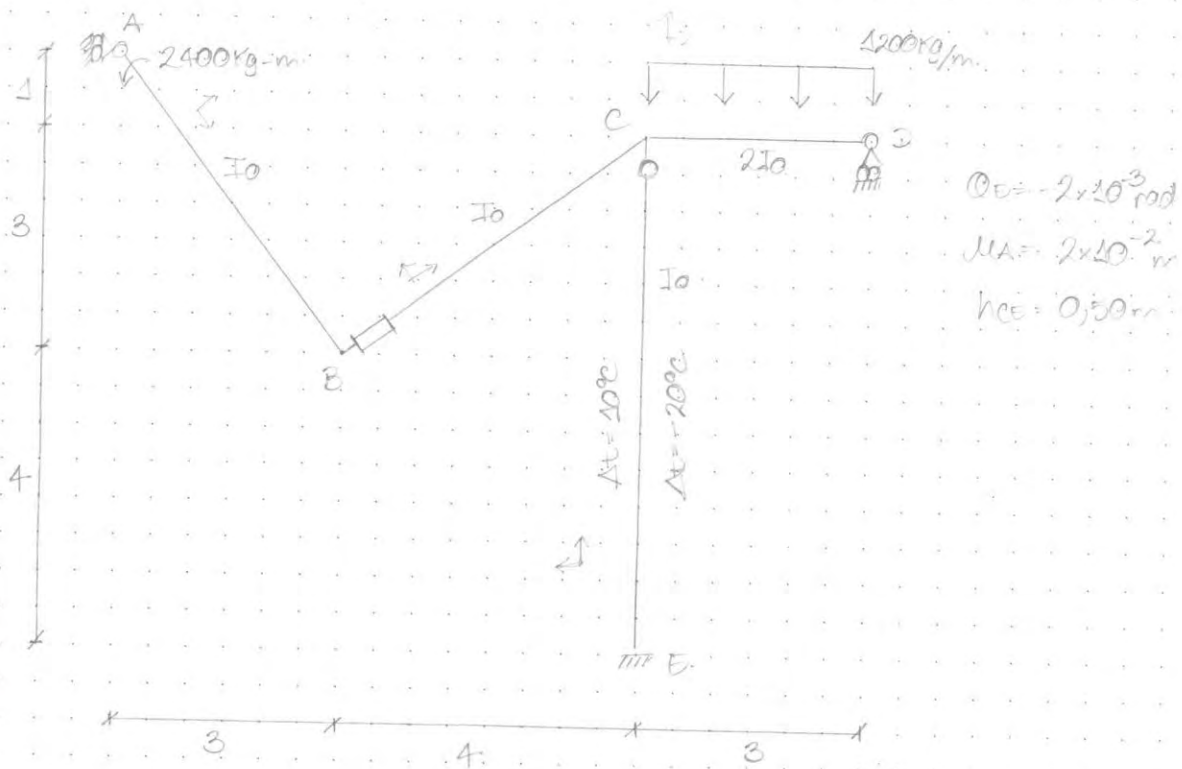
$$+ \frac{3}{6 \times 2EI_0} [2 \times (-3600) \times (-10) + 2 \times (-11700) \times (-10) + (-10) \times (-11700 - 3600)]$$

$$+ \frac{10^6}{EI_0} \times \frac{10^{-5}}{0,50} \times (-10) \times (-10) \times 3 + \frac{2 \times (-10) \times 675 \times 3}{3 \times 2EI_0}$$

$$S = \frac{406700}{EI_0}$$

PREPARADOR: RODRIGUEZ J. VÍCTOR

3) Calcular el trabajo: $S = 3Qb - VA + MD$



$$1) \sum M_B^{\text{ext}} = 0 \Rightarrow 2400 + M_{BA} = 0 \Rightarrow M_{BA} = -2400 \text{ kg-m}$$

$$2) \sum M_B^{\text{int}} = 0 \Rightarrow M_{BA} + 2400 - A_h \times 4 = 0 \Rightarrow A_h = 0$$

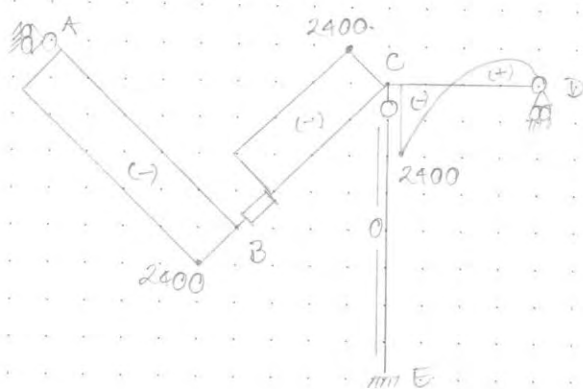
$$3) \sum \vec{F}_h = 0 \Rightarrow A_h + E_h = 0 \Rightarrow E_h = 0$$

$$4) \sum M_E^{\text{ext}} = 0 \Rightarrow E_h \times 4 - M_E = 0 \Rightarrow M_E = 0$$

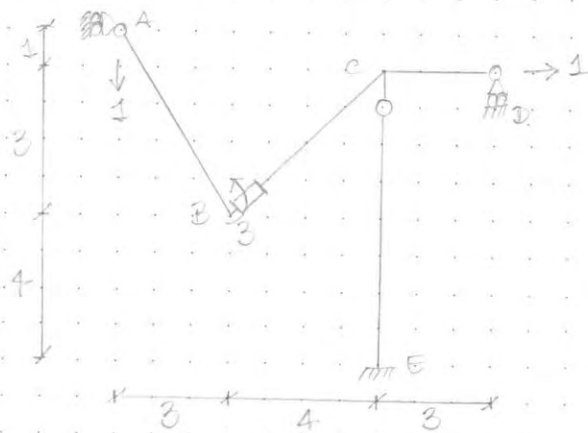
$$5) \sum M_E^{\text{int}} = 0 \Rightarrow -M_E + 2400 - A_h \times 8 - 1200 \times 3 \times \frac{3}{2} + D \times 3 = 0 \Rightarrow D = 1000 \text{ kg}$$

$$6) \sum M_D^{\text{ext}} = 0 \Rightarrow -M_{CD} + D \times 3 - 1200 \times 3 \times \frac{3}{2} = 0 \Rightarrow M_{CD} = -2400 \text{ kg-m}$$

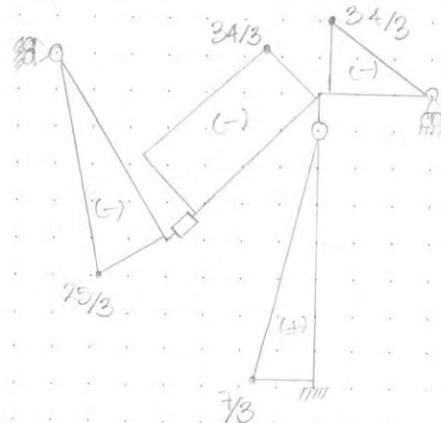
- Diagrama de Momentos Sistema P



- Sistema P'



- Diagrama de Momentos Sistema P'



$$1) \sum \overset{\curvearrowright}{M}_A = 0 \Rightarrow M_{BA} + 1 \times (7 + 4/3) = 0 \Rightarrow M_{BA} = -25/3$$

$$2) \sum \overset{\curvearrowright}{M}_B = 0 \Rightarrow M_{BA} + 1 \times 3 - A_h \times 4 = 0 \Rightarrow A_h = -1/3$$

$$3) \sum F_h = 0 \Rightarrow A_h + 1 + E_h = 0 \Rightarrow E_h = -1/3$$

$$4) \sum \overset{\curvearrowright}{M}_E = 0 \Rightarrow E_h \times 7 - M_E = 0 \Rightarrow M_E = 7/3$$

$$5) \sum \overset{\curvearrowright}{M}_C = 0 \Rightarrow -1 \times 7 + D_v \times 3 + 3 + 1 \times 7 - A_h \times 8 - M_E = 0 \Rightarrow D_v = -34/3$$

$$6) \sum F_v = 0 \Rightarrow D_v - E_v - 1 = 0 \Rightarrow E_v = -43/3$$

$$7) \sum \overset{\curvearrowright}{M}_D = 0 \Rightarrow -M_D + D_v \times 3 = 0 \Rightarrow M_D = -34/3$$

PREPARADOR PEDRO GARCIA - VIREI 2

Proyecto TEMA 2- APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES.

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$$\begin{aligned} S + 2 \times 10^{-3} \times 7 \times \frac{10^6}{EI_0} - 2 \times 10^{-2} \times \frac{4}{3} \times \frac{10^6}{EI_0} &= \frac{(-2400)(1-\frac{25}{3}) \times 5}{2EI_0} + \frac{(-2400) \times (-\frac{34}{3}) \times 5}{EI_0} \\ &+ \frac{(-2400) \times (-\frac{24}{3}) \times 3}{3 \times 2EI_0} + \frac{1850 \times (-\frac{34}{3}) \times 3}{3 \times 2EI_0} \\ &+ \frac{10^5 \times 10^6 \times (-5)(-\frac{43}{2}) \times 7}{EI_0} + \frac{10^5 \times 10^6 \times \frac{7}{3} \times 7 \times (-30)}{0,50 \times EI_0 \times 2} \end{aligned}$$

$$S = \frac{210722,22}{EI_0}$$

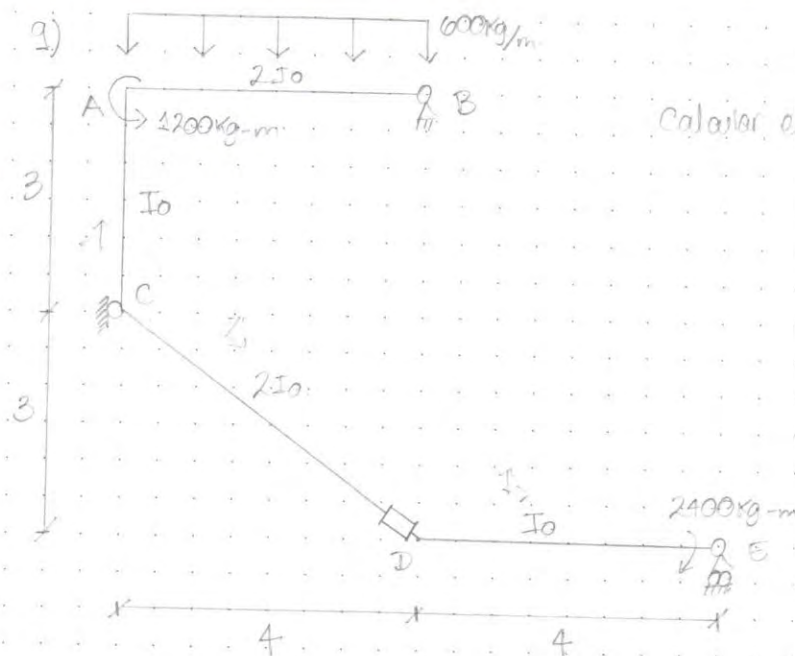
PREPAREDOR: REIDENAR T. VILALBA

TEMA 2- APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL PARA CUERPOS DEFORMABLES

Proyecto

Fecha 0010/08/2022

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Calcular el trabajo $\Delta = 2u_E - 10^4 \theta_C - 3v_C$

$$u_E = -2 \times 10^{-2} \text{ m}$$

$$\Delta t_0 = 10^\circ \text{ C}$$

$$\alpha t = 10^{-5} \text{ m}^2$$

$$EI_0 = 10^6 \text{ kg-m}^2$$

$$1) \sum M_D = 0 \Rightarrow -M_D - 2400 = 0 \Rightarrow M_D = -2400 \text{ kg-m}$$

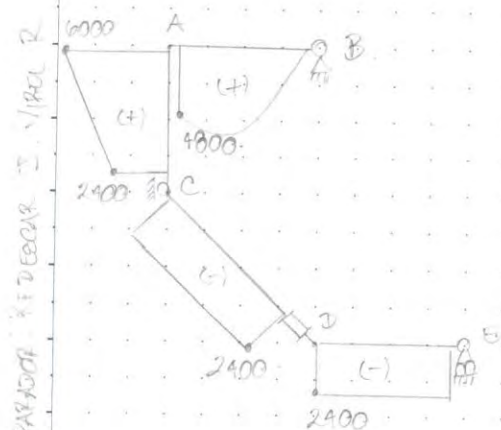
$$2) \sum M_D = 0 \Rightarrow -M_D - 2400 + E \times 4 = 0 \Rightarrow E = 0$$

$$3) \sum M_C = 0 \Rightarrow C \times 3 + 1200 + 600 \times 4 \times 2 = 2400 + E \times 4 = 0 \Rightarrow C = -1200 \text{ kg}$$

$$4) \sum M_C = 0 \Rightarrow -M_C - 2400 + E \times 8 = 0 \Rightarrow M_C = -2400 \text{ kg-m}$$

$$5) \sum M_A = 0 \Rightarrow M_A + C \times 3 - 2400 + E \times 8 = 0 \Rightarrow M_A = 6000 \text{ kg-m}$$

Diagrama de Momentos Sistema P



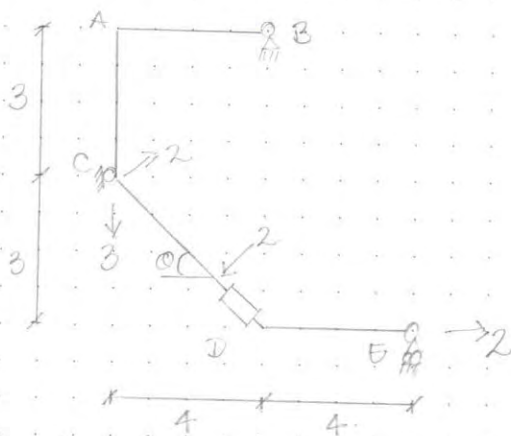
TEMA 2 - APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES

Proyecto

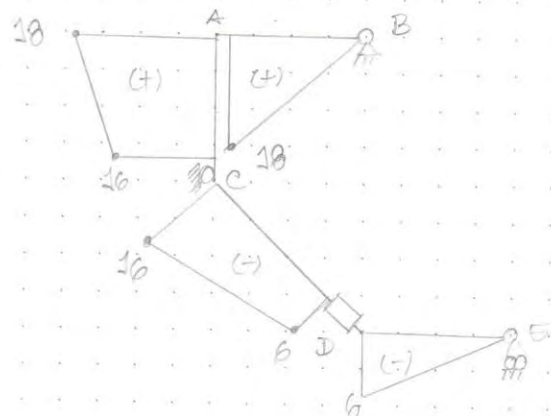
Fecha 02/09/2022

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- Sistema P



- Diagrama de Momentos Sistema P



$$1) \sum M_D^+ = 0 \Rightarrow -M_{DE} - 2 \times 3 = 0 \Rightarrow M_{DE} = -6 \text{ kg-m}$$

$$2) \sum M_D^+ = 0 \Rightarrow -M_{DE} + E_N \times 4 = 0 \Rightarrow E_N = -1.5 \text{ kg}$$

$$3) \sum F_V = 0 \Rightarrow B_V + E_N - 2 = 0 \Rightarrow B_V = 4.5 \text{ kg}$$

$$4) \sum M_A^+ = 0 \Rightarrow -M_{AB} + B_V \times 4 = 0 \Rightarrow M_{AB} = 18 \text{ kg-m}$$

$$5) \sum M_C^+ = 0 \Rightarrow -M_{CD} - 2 \times 5 + E_N \times 8 + 2 \times 3 = 0 \Rightarrow M_{CD} = -16$$

$$6) \sum F_H = 0 \Rightarrow 2 - N_{CD} = 0 \Rightarrow N = 2.5 \text{ kg (Fuerza Axial en CD)}$$

$$7) \sum M_P^+ = 0 \Rightarrow B_V \times 8 + M_{DE} + 2 \times 5 - 2 \times 30 - 3 \times 4 + C_H \times 3 = 0 \Rightarrow C_H = -8/3$$

$$S + 2 \times 10^{-2} \times \frac{8}{3} \times 10^6 = \frac{4800 \times 18 \times 4}{3 \times 2500} + \frac{1200 \times 18 \times 4}{6 \times 10} + \frac{3}{6 \times 10} [2 \times 2400 \times 16 + 2 \times 6000 \times 16 + 16 \times 6000 + 18 \times 1400] \\ + \frac{16}{6 \times 2500} [2 \times (-16) \times (-2400) + 2 \times (-16) \times (-2400) + (-2400) \times (-6 - 16)] + \frac{(-6) \times (-2400) \times 4}{2500} \\ + 10^{-5} \times \frac{10^6}{10} \times 10 \times 2.5 \times 5$$

$$S = 3307.16167 / \text{EIO}$$

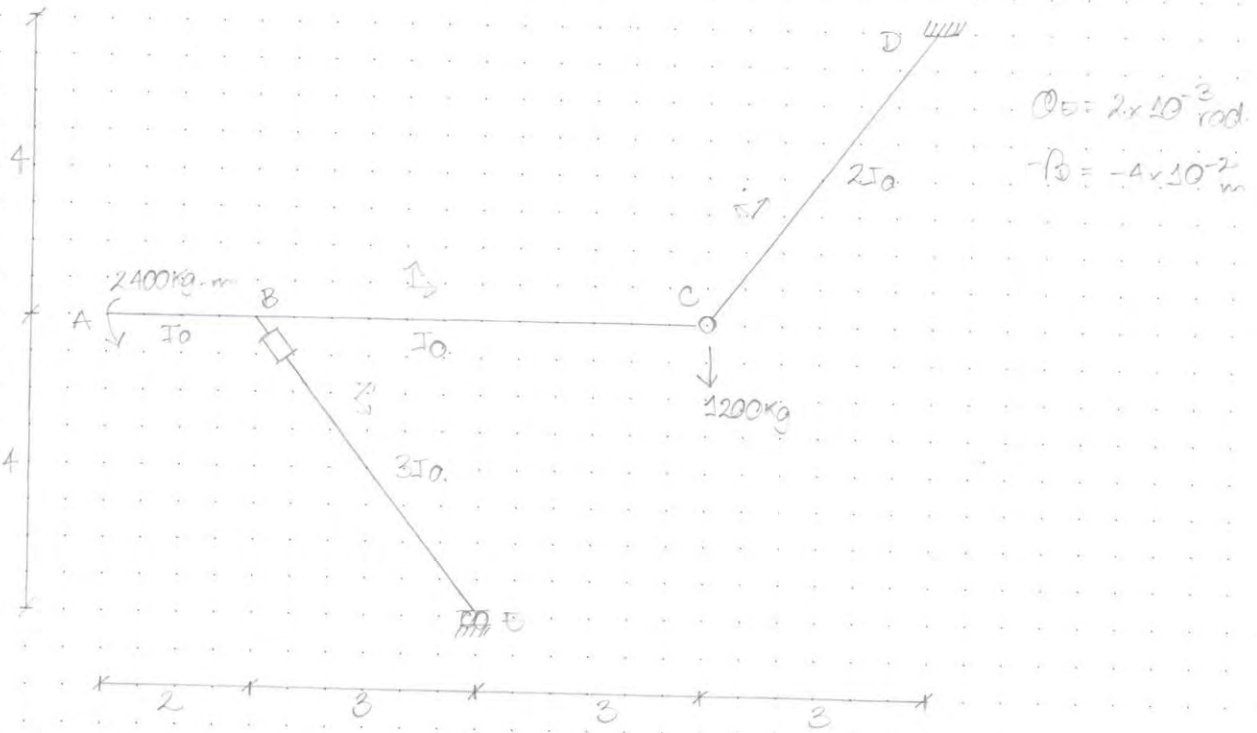
Preparado: REDONAR T. VARELA

TEMA 2-APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
TAPA CUERPOS DEFORMABLES

Fecha OCTUBRE 2012

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10) Halle las ecuaciones de la elástica.



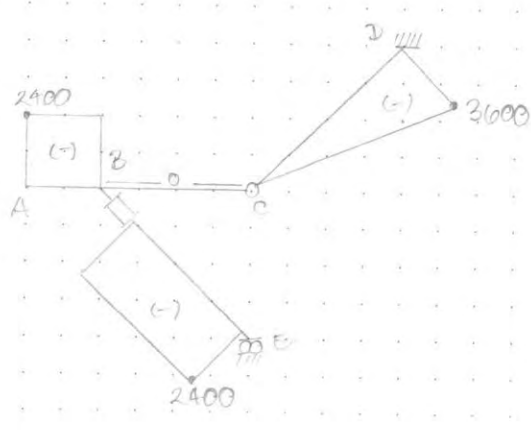
$\theta_D = 2 \times 10^{-3} \text{ rad}$
 $\theta_B = -4 \times 10^{-2} \text{ rad}$

1) $\sum M_B^+ = 0 \Rightarrow -M_A + M_E + E_V \times 3 = 0 \Rightarrow E_V = 0$

2) $\sum M_C^+ = 0 \Rightarrow 2400 - E_V \times 3 + M_E = 0 \Rightarrow M_E = -2400 \text{ Kg-m}$

3) $\sum M_D^+ = 0 \Rightarrow M_D + 2400 + 1200 \times 3 - E_V \times 6 + M_E = 0 \Rightarrow M_D = -3600 \text{ Kg-m}$

-Diagrama de Momentos



PROFESOR REYES J. 1999.7

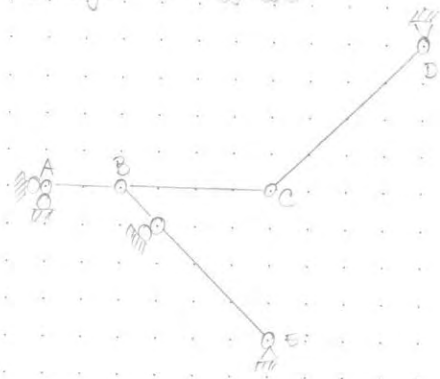
TEMA 2 - APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES

Proyecto

Fecha OCTUBRE 2012

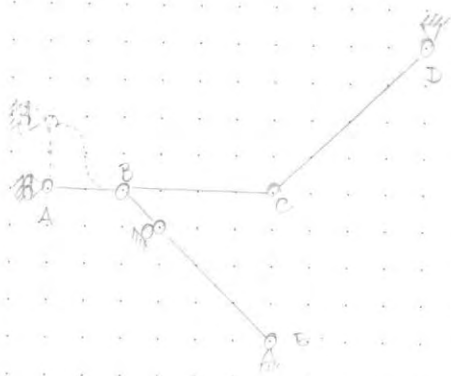
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Imagen animática

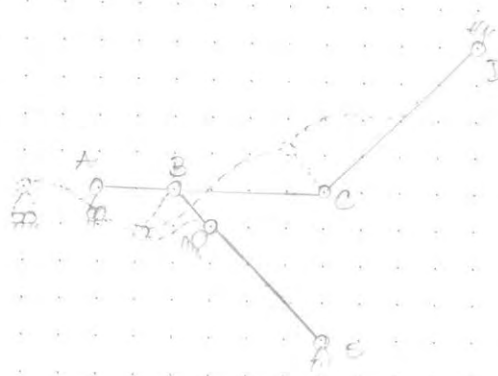


Grados de Desplazabilidad = 3

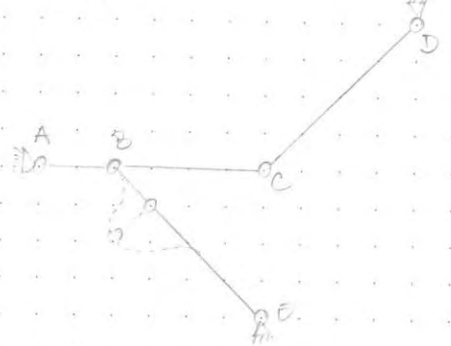
Modo I



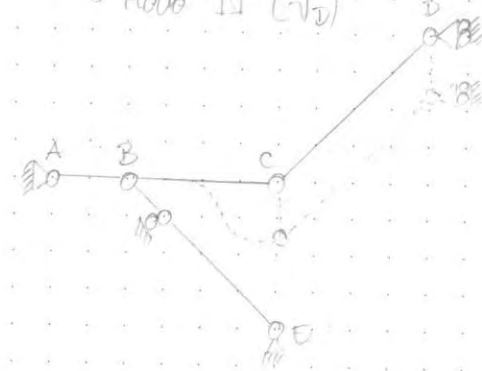
Modo II



Modo III



Modo III (V_D)



PROYECTO RESOLUCIÓN I. / REL 2

TEMA 2- APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL PARA CUERPOS DEFORMABLES

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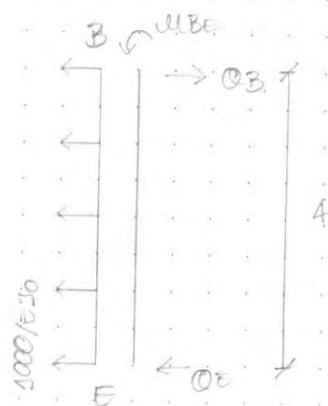
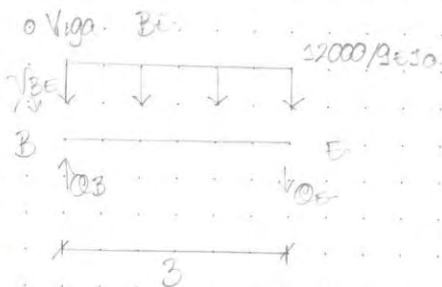
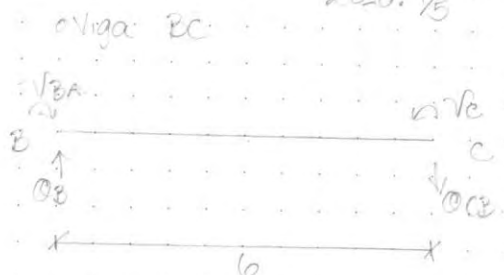
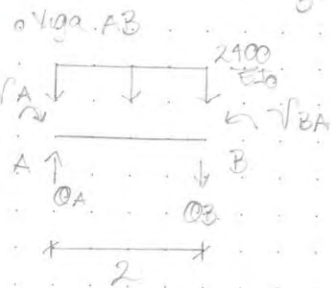
JUNTA MODO	A		B		C		D		E	
I	0	V_A	0	0	0	0	0	0	0	0
II	M_A	0	M_{BA}	V_{BA}	M_C	V_C	0	0	0	0
III	0	0	M_{BE}	V_{BE}	0	0	0	0	0	0
IV	0	0	0	0	0	V_C	0	V_D	0	0
SUPERFICIE	M_A	V_A	M_{BA} M_{BE}	V_{BA} V_{BE}	M_C	V_C	0	V_D	0	0
Q	Q_A		Q_B		Q_C, Q_D		0		Q_E	

$$V_D = -\frac{40000}{EJ_0}$$

$$Q_E = \frac{2000}{EJ_0}$$

$$q_{AB}^V = -\frac{2400}{EJ_0}; \quad q_{BE}^V = -\frac{2400}{3EJ_0 \cdot \frac{3}{5}} = -\frac{22000}{9EJ_0}; \quad q_{BE}^H = -\frac{2400}{3EJ_0 \cdot (-\frac{4}{5})} = \frac{4000}{EJ_0}$$

$$q_{CD}^V = -\frac{3600}{2EJ_0 \cdot \frac{3}{5}} = -\frac{3000}{EJ_0}; \quad q_{CD}^H = -\frac{3600}{2EJ_0 \cdot \frac{4}{5}} = -\frac{2250}{EJ_0}$$

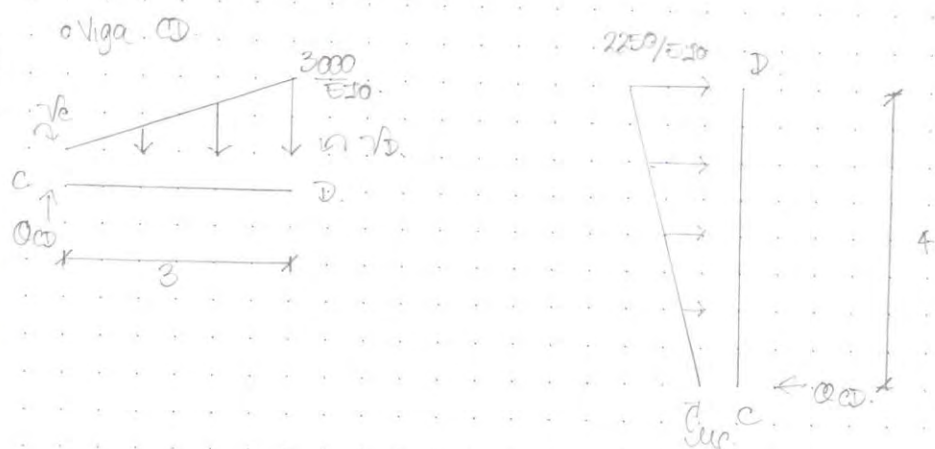


PREPARED: REVISOR: J. VIREL P.

TEMA 2-APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL PARA CUERPOS DEFORMABLES

Proyecto

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o Viga Horizontal CD

$$\sum F_v = 0 \Rightarrow Q_C - \frac{3000}{EI_0} \times \frac{3}{2} = 0 \Rightarrow Q_C = \frac{4500}{EI_0}$$

$$\sum M_C^D = 0 \Rightarrow -V_C - Q_C \times 3 + V_D + \frac{3000}{EI_0} \times \frac{3}{2} \times 1 = 0 \Rightarrow V_C = -\frac{42000}{EI_0}$$

o Viga Vertical CD

$$\sum M_C^D = 0 \Rightarrow -M_C - \frac{2250}{EI_0} \times \frac{4}{2} \times \frac{2}{3} \times 4 = 0 \Rightarrow M_C = -\frac{12000}{EI_0}$$

o Viga Horizontal BE

$$\sum F_v = 0 \Rightarrow Q_B - Q_C - \frac{12000}{EI_0} \times 3 = 0 \Rightarrow Q_B = \frac{6000}{EI_0}$$

$$\sum M_B^E = 0 \Rightarrow -V_B - Q_B \times 3 - \frac{12000}{EI_0} \times 3 \times \frac{3}{2} = 0 \Rightarrow V_B = -\frac{12000}{EI_0}$$

o Viga Vertical BE

$$\sum M_B^E = 0 \Rightarrow M_B - \frac{12000}{EI_0} \times 4 \times 2 - Q_B \times 4 = 0 \Rightarrow M_B = \frac{16000}{EI_0}$$

o Viga BC

$$\sum F_v = 0 \Rightarrow Q_C = Q_B \Rightarrow Q_C = \frac{6000}{EI_0}$$

$$\sum M_C^B = 0 \Rightarrow V_C - Q_B \times 6 - V_B = 0 \Rightarrow V_C = -85000/EI_0$$

o Viga AB

$$\sum F_v = 0 \Rightarrow Q_A - Q_B - \frac{2400}{EI_0} \times 2 = 0 \Rightarrow Q_A = \frac{10800}{EI_0}$$

$$\sum M_A^B = 0 \Rightarrow -V_A + V_B - Q_B \times 2 - \frac{2400}{EI_0} \times 2 = 0 \Rightarrow V_A = -101800/EI_0$$

Principio de Trabajo Virtual

- Ecuaciones de la Elástica

- viga AB

$$q(x) = \frac{-2400}{EI_0}$$

$$Q(x) = \frac{1}{EI_0} (-2400x + 108000)$$

$$V(x) = \frac{1}{EI_0} (-1200x^2 + 108000x - 108000)$$

- viga BC

$$q(x) = \frac{6000}{EI_0}$$

$$V(x) = \frac{1}{EI_0} (6000x - 35000)$$

- viga BE

- Elástica Vertical

$$q(x) = \frac{-12000}{EI_0}$$

$$Q(x) = \frac{1}{EI_0} (-12000x + 6000)$$

$$V(x) = \frac{1}{EI_0} (-6000x^2 + 6000x - 12000)$$

- Elástica Horizontal

$$q(x) = \frac{1000}{EI_0}$$

$$Q(x) = \frac{1}{EI_0} (1000x + 2000)$$

$$V(x) = \frac{1}{EI_0} (500x^2 + 2000x)$$

- viga CD

- Elástica Horizontal

$$q(x) = \frac{-562,5x}{EI_0}$$

$$Q(x) = \frac{1}{EI_0} (-281,25x^2 + 4500)$$

$$V(x) = \frac{1}{EI_0} \left(-\frac{281,25}{3}x^3 + 4500x - 12000 \right)$$

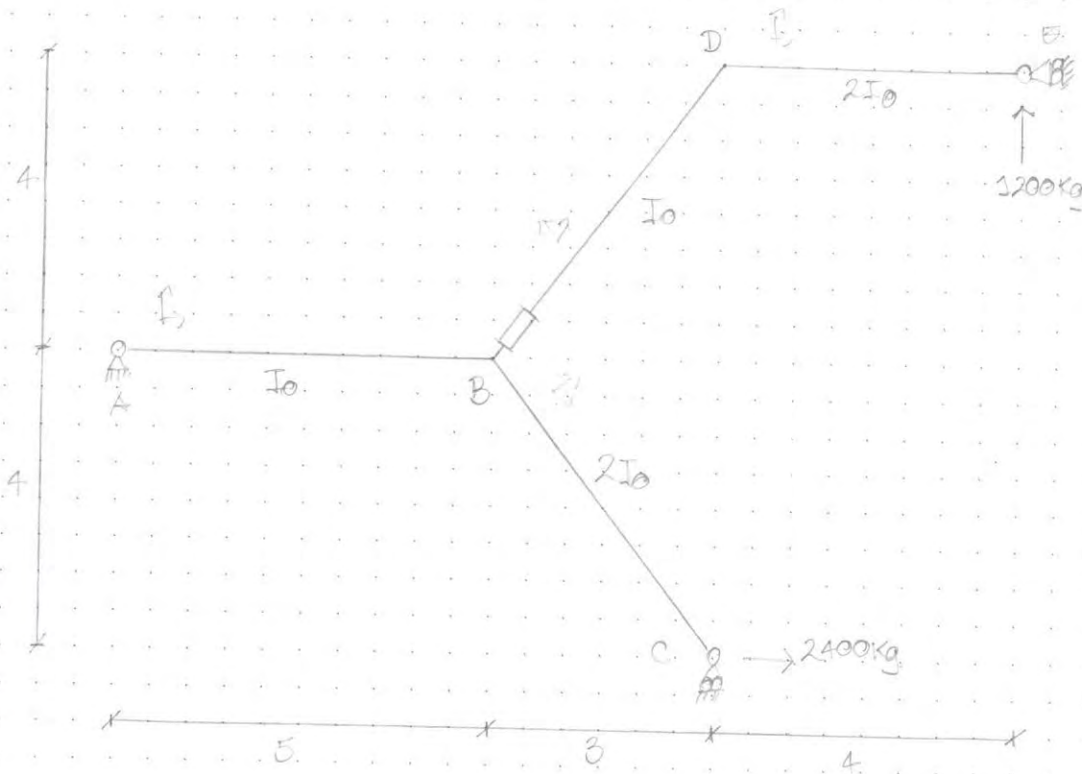
- Elástica Vertical

$$q(x) = \frac{-4000x}{EI_0}$$

$$Q(x) = \frac{1}{EI_0} (-2000x^2 + 4500)$$

$$V(x) = \frac{1}{EI_0} \left(-\frac{2000}{3}x^3 + 4500x - 42000 \right)$$

11) Calcular el trabajo $S = 30E - u_{ic} + 27E$



$$1) \sum M_D^D = 0 \Rightarrow -M_{DE} + 1200 \times 4 = 0 \Rightarrow M_{DE} = 4800 \text{ kg-m}$$

$$2) \sum M_B^{D+} = 0 \Rightarrow -4800 + 1200 \times 7 - E_h \times 4 = 0 \Rightarrow E_h = 900 \text{ kg}$$

$$3) \sum M_A^{int+} = 0 \Rightarrow 2400 \times 4 + 1200 \times 12 - E_h \times 4 + C_v \times 8 = 0 \Rightarrow C_v = -2550 \text{ kg}$$

$$4) \sum M_E^{D+} = 0 \Rightarrow -M_{BE} + 2400 \times 4 + C_v \times 3 = 0 \Rightarrow M_{BE} = 1950$$

TEMA 2- APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES

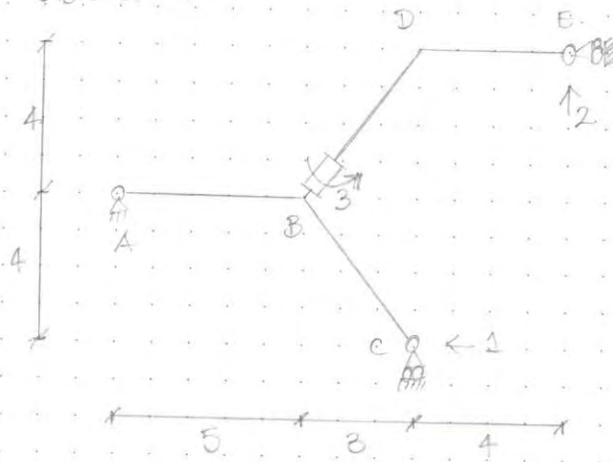
Proyecto

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- Sistema P'



$$1) \sum M_D^{\text{ext}} = 0 \Rightarrow -M_{DE} + 2 \times 4 = 0 \Rightarrow M_{DE} = 8 \text{ kg-m}$$

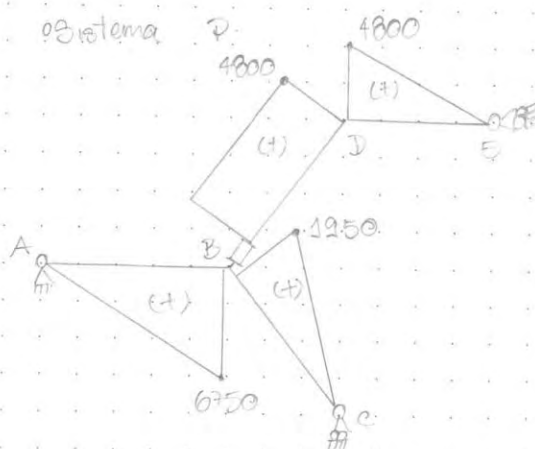
$$2) \sum M_B^{\text{ext}} = 0 \Rightarrow -8 + 2 \times 7 - E_h \times 4 = 0 \Rightarrow E_h = 1.5 \text{ kg}$$

$$3) \sum M_A^{\text{ext}} = 0 \Rightarrow 3 + 2 \times 12 - 1 \times 4 + C_v \times 8 - E_h \times 4 = 0 \Rightarrow C_v = -17/8 \text{ kg}$$

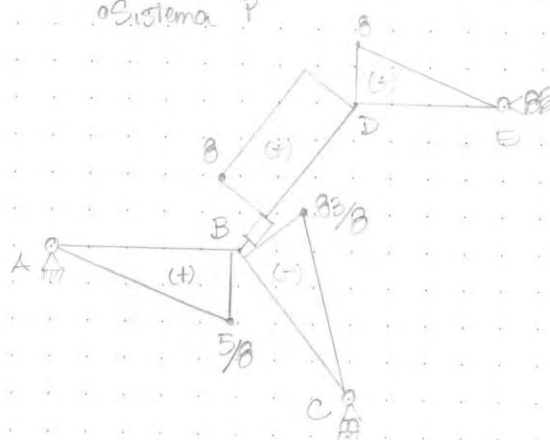
$$4) \sum M_B^{\text{ext}} = 0 \Rightarrow -M_{BC} - 1 \times 4 + C_v \times 3 = 0 \Rightarrow M_{BC} = -83/8 \text{ kg-m}$$

- Diagramas de Momento

o Sistema P'



o Sistema P'



PREPARADOR: REYES R. / 19/09/12

Proyecto TEMA 2- APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES

Fecha OCTUBRE 2012

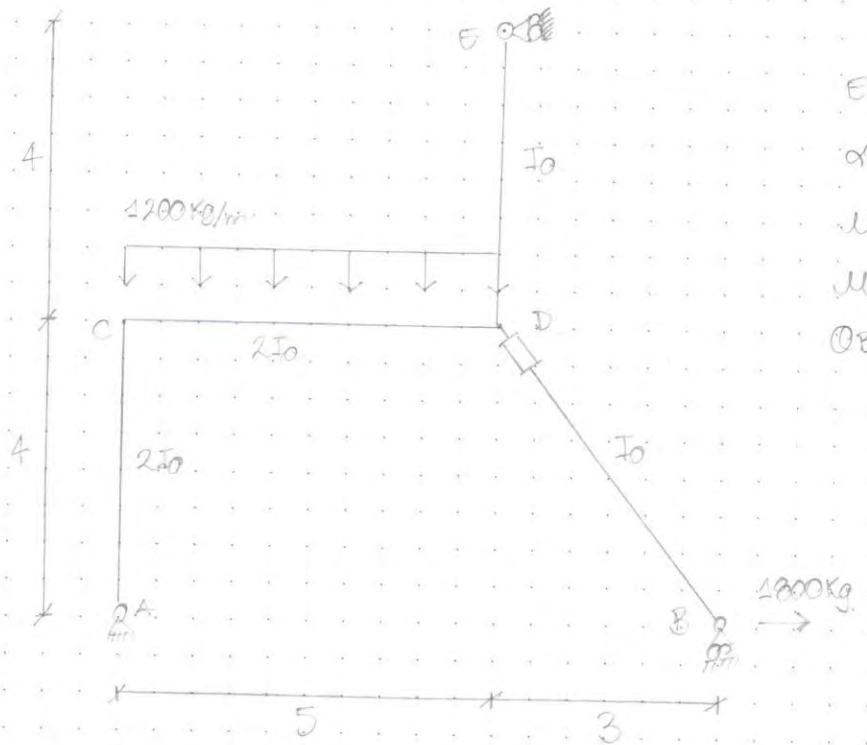
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$$S = \frac{6750 \times (5\%) \times 5}{3EIo} + \frac{1250 \times (-8\%) \times 5}{3 \times 2EIo} + \frac{4000 \times 8 \times 5}{EIo} + \frac{4000 \times 8 \times 4}{3 \times 2EIo}$$

$$S = 207.772 / EIo$$

PREPARADOR: REJESDAE J. VIREA R

12.) Defina las Ecuaciones de la Elástica



$$EI_0 = 10^6 \text{ kg-m}^2$$

$$\alpha = 10^{-5} \text{ rad}^{-1}$$

$$M_A = -3 \times 10^{-2} \text{ m}$$

$$M_B = 2 \times 10^{-2} \text{ m}$$

$$\theta_B = 6 \times 10^{-3} \text{ rad}$$

$$1) \sum M_D^{\circ} = 0 \Rightarrow -M_{DB} + 1800 \times 4 + B_v \times 3 = 0 \Rightarrow B_v = -2400 \text{ kg}$$

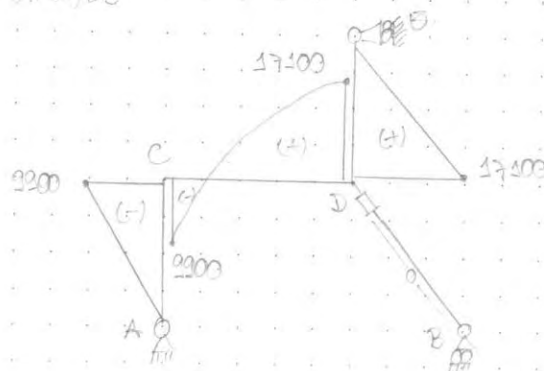
$$2) \sum M_A^{\circ} = 0 \Rightarrow -E_h \times 8 - 1200 \times 5 \times \frac{5}{2} + B_v \times 8 = 0 \Rightarrow E_h = -4275 \text{ kg}$$

$$3) \sum F_h^{\rightarrow} = 0 \Rightarrow A_h + 1800 + E_h = 0 \Rightarrow A_h = 2475 \text{ kg}$$

$$4) \sum M_D^{\circ} = 0 \Rightarrow -M_{DE} - E_h \times 4 = 0 \Rightarrow M_{DE} = 17100 \text{ kg-m}$$

$$5) \sum M_C^{\circ} = 0 \Rightarrow M_{CA} + A_h \times 4 = 0 \Rightarrow M_{CA} = -9900 \text{ kg-m}$$

-Diagrama de Momentos



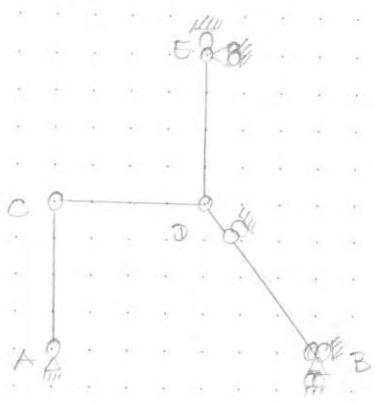
TEMA 2 - APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES

Proyecto

Fecha 00/08/25 20:22

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-Imagen Cinemática

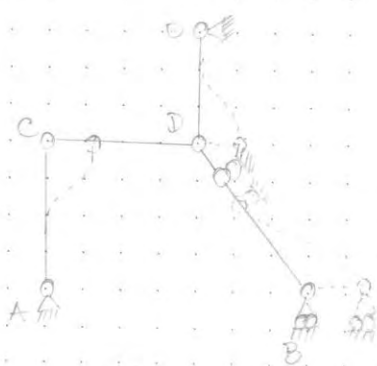


Grados de Libertad = 3

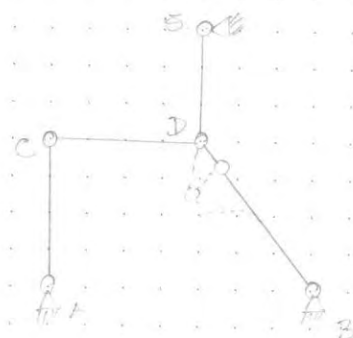
Modo I



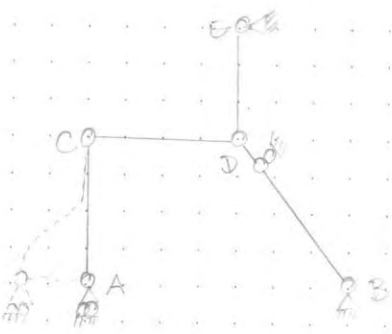
Modo II



Modo III



Modo IV (U.A)



PREPARADOR: REYES J. YRUELA

Proyecto TEMA 2-APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL PARA CUERPOS DEFORMABLES.

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JUNTA NODO	A		B		C		D		E	
I	0	0	0	0	M_C	0	M_D	V_D	0	V_E
II	0	0	M_B	0	M_C	0	M_D	M_{DB}	0	0
III	0	0	0	0	0	0	M_D	V_D	0	0
IV	M_A	0	0	0	0	0	0	0	0	0
SUPERFICIÓN	M_A	0	M_B	0	M_C	0	M_D	V_D	0	V_E
Q	Q_A		Q_B		Q_C		Q_D		Q_E	

$$M_A = 30000/EI_0$$

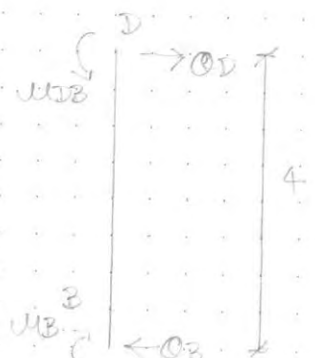
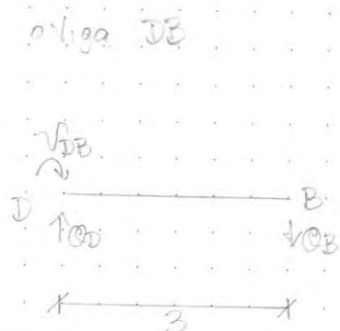
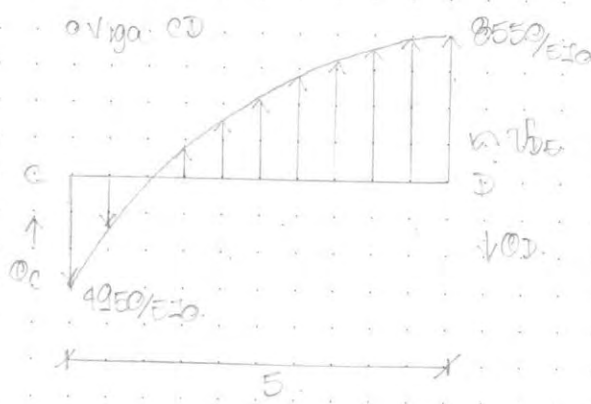
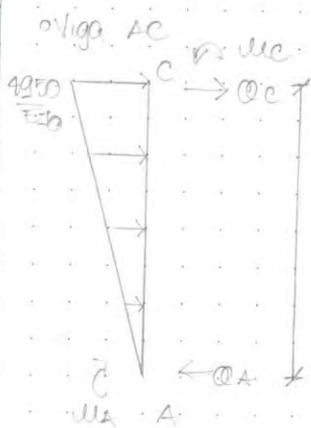
$$M_B = -20000/EI_0$$

$$Q_E = 6000/EI_0$$

Viga Conjugada

$$q_{AC}^h = -\frac{9200}{25EI_0} = -\frac{4250}{EI_0}; \quad q_{DE}^h = \frac{17100}{EI_0}$$

$$q_{AC}^v = -\frac{9200}{25EI_0} = -\frac{4250}{EI_0}; \quad q_{DE}^v = \frac{17100}{25EI_0} = \frac{6850}{EI_0}$$



PREPARAR REDESAR 3. VUELTA

• Viga DB Vertical

$$\sum F_v = 0 \Rightarrow -Q_B + Q_D = 0 \Rightarrow Q_D = 6000/\text{Eso}$$

$$\sum M_D^+ = 0 \Rightarrow 1500 - Q_B \times 4 - M_B = 0 \Rightarrow M_{DB} = 4000/\text{Eso}$$

• Viga DB Horizontal

$$\sum M_B^+ = 0 \Rightarrow -V_{DB} - Q_B \times 3 = 0 \Rightarrow V_{DB} = -18000/\text{Eso}$$

• Viga DE

$$\sum F_v = 0 \Rightarrow -Q_D - \frac{17100}{\text{Eso}} \times \frac{4}{2} + Q_E = 0 \Rightarrow Q_E = 40200/\text{Eso}$$

$$\sum M_D^+ = 0 \Rightarrow -M_{DE} + \frac{17100}{\text{Eso}} \times \frac{4}{2} \times \frac{1}{3} \times 4 - Q_E \times 4 = 0 \Rightarrow M_{DE} = -\frac{115200}{\text{Eso}}$$

• Viga CD

$$\sum F_v = 0 \Rightarrow Q_C - Q_D - \frac{1950}{\text{Eso}} \times \frac{5}{2} + \frac{8550}{\text{Eso}} \times \frac{5}{2} + \frac{2}{3} \times 5 \times 3750 = 0 \Rightarrow Q_C = -\frac{15500}{\text{Eso}}$$

$$\sum M_C^+ = 0 \Rightarrow -Q_D \times 5 - \frac{1950}{\text{Eso}} \times \frac{5}{2} \times \frac{1}{3} \times 5 + \frac{8550}{\text{Eso}} \times \frac{5}{2} \times \frac{2}{3} \times 5 + \frac{2}{3} \times 5 \times 3750 \times \frac{5}{2} + M_{DE} = 0$$

$$M_{DE} = -\frac{51875}{\text{Eso}}$$

• Viga AC

$$\sum F_v = 0 \Rightarrow -Q_A + Q_C + \frac{1950}{\text{Eso}} \times \frac{4}{2} = 0 \Rightarrow Q_A = -5600/\text{Eso}$$

$$\sum M_A^+ = 0 \Rightarrow -M_A - \frac{1950}{\text{Eso}} \times \frac{4}{2} \times \frac{2}{3} \times 4 - Q_C \times 4 + M_{DE} = 0 \Rightarrow M_{AC} = -5600/\text{Eso}$$

TEMA 2- APLICACIONES DEL PRINCIPIO DE TRABAJO VIRTUAL
PARA CUERPOS DEFORMABLES

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-Ecuaciones de la Elástica

oViga AC

$$Q(x) = -1237,5x$$

$$Q(x) = \frac{1}{E_b} \left(-\frac{1237,5}{2} x^2 - 5000 \right)$$

$$M(x) = \frac{1}{E_b} \left(-\frac{1237,5}{6} x^3 - 5000x + 30000 \right)$$

oViga CD

$$Q(x) = -1200$$

$$V_y(x) = -1200x + 8400$$

$$M_z(x) = -600x^2 + 8400x - 2300$$

$$Q(x) = \frac{1}{E_b} \left(-200x^3 + 4200x^2 - 2200x - 15500 \right)$$

$$T(x) = \frac{1}{E_b} \left(-50x^4 + 4200x^3 - 4950x^2 - 15500x \right)$$

oViga DE

$$Q(x) = -4275x + 17100$$

$$Q(x) = \frac{1}{E_b} \left(-\frac{4275}{2} x^2 + 17100x + 6000 \right)$$

$$M(x) = \frac{1}{E_b} \left(-\frac{4275}{6} x^3 + \frac{17100}{2} x^2 + 6000x - 135200 \right)$$

oViga DB

-Elástica vertical

$$Q(x) = \frac{6000}{E_b}$$

$$V(x) = \frac{1}{E_b} (6000x - 18000)$$

-Elástica horizontal

$$Q(x) = \frac{6000}{E_b}$$

$$M(x) = \frac{1}{E_b} (6000x - 20000)$$

PREPARADOR REDESAR 3. VÍDEO 2